Canonical Symmetric–Exterior Spectral Decomposition, Cubic Moment Reduction, and Partial Resolution of Rossman's Conjectures on Sidorenko's Conjecture for Bipartite Möbius Ladders ($K_{5,5} \setminus C_{10}$)

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Abstract

Sidorenko's Conjecture (1991) asserts that for every bipartite graph H and every symmetric graphon W (equivalently, every nonnegative symmetric $n \times n$ matrix G), the homomorphism density of H in W is at least that in a uniform random graph of the same edge density. Since the foundational work of Sidorenko (1991) and Hatami (2010), the 10-vertex graph $K_{5,5} \setminus C_{10}$—isomorphic to the 5-rung bipartite Möbius ladder ML_{10}—has stood as the smallest and most prominent open test case of Sidorenko's Conjecture. Recently, Rossman (2025) proved that for every m-rung bipartite ladder BL_{2m}, $\hom(BL_{2m}, G) = \mathrm{Tr}(\widetilde{G}^m)$ where $\widetilde{G}_{ij,kl} = G_{il}G_{kj}\sqrt{G_{ij}G_{kl}} = (D(G \otimes G)PD)_{ij,kl}$ is an $n^2 \times n^2$ twisted tensor operator, and embedded $\widetilde{G}$ at $t = 1/3$ into the continuous one-parameter family $M(t)_{ij,kl} = (A_{ij}A_{kl})^t(A_{il}A_{kj})^{1-t}$ ($A = G^{\circ 3/2}$, $t \in [0,1]$). Rossman conjectured that $\widetilde{G}$ has a *pseudo-positive spectrum* (Conjecture 3.2, implying Sidorenko's Conjecture for $K_{5,5} \setminus C_{10}$ and all odd-m Möbius ladders ML_{2m}), satisfies the pairwise eigenvalue majorization $(\widetilde{\lambda}_2 + \cdots + \widetilde{\lambda}_k) + (\widetilde{\lambda}_{n^2-k+2} + \cdots + \widetilde{\lambda}_{n^2}) \geq 0$ (Conjecture $(**)$), and more generally that $M(t)$ is pseudo-positive for *all* $t \in [0,1]$ (Conjecture 4.2).

In this paper, we establish six theorems that expose the exact algebraic and spectral structure of $\widetilde{G}$ and $M(t)$ and push the unconditional resolution of Rossman's conjectures beyond rank 2:

(1) **Canonical Symmetric–Exterior Orthogonal Decomposition, Universal Trace Identity, & Extended Star-Subspace Identity (Theorems 2.1 and 2.3):** For every symmetric $n \times n$ matrix A and every $t \in [0,1]$, $M(t) = \mathcal{L}_+(t) \oplus (-\mathcal{L}_-(t))$ on $\mathbb{R}^{n\times n} = \mathrm{Sym}_n(\mathbb{R}) \oplus^{\perp} \mathrm{Skew}_n(\mathbb{R})$, with $\mathrm{Tr}(\mathcal{L}_+(t) - ww^T) = \mathrm{Tr}(\mathcal{L}_-(t))$ and the universal exterior-trace identity $\langle V, (\mathrm{Sym}^2 B - \mathrm{vec}(B)\,\mathrm{vec}(B)^T)V\rangle = -2\,\mathrm{Tr}((\wedge^2 V)(\wedge^2 B))$. Moreover, on the n-dimensional extended star subspace $\widehat{\mathcal{S}}_v^+ = \mathbb{R}E_{vv} \oplus \mathrm{span}\{E_{vi}^+ : i \neq v\}$ of any vertex $v \in \{1,\ldots,n\}$, $(\mathcal{L}_+(t) - ww^T)|_{\widehat{\mathcal{S}}_v^+} = 0_1 \oplus \mathcal{L}_-(t)|_{\mathcal{S}_v^-}$.

(2) **Square-Root-Free Cubic Moment Tensor Reduction (Theorem 3.1):** By similarity, $\mathrm{Spec}(\widetilde{G}) = \mathrm{Spec}(D^2(G \otimes G)P)$, eliminating all fractional powers from the characteristic polynomial of $\widetilde{G}$. For any symmetric matrix $G = \sum_{e=1}^r \varepsilon_e x^{(e)}(x^{(e)})^T$ of rank r and arbitrary dimension $n \times n$, the nonzero spectrum of $\widetilde{G}$ coincides with that of the $r^2 \times r^2$ cubic moment operator $\mathcal{M}_r = \mathcal{M}_r^+ \oplus (-\mathcal{N}_r^-)$, where $(S_e)_{ab} = \sum_{i=1}^n x_i^{(e)} x_i^{(a)} x_i^{(b)}$.

(3) **Unconditional Resolution for All Rank-≤ 2 Matrices and Graphons (Theorem 4.3):** Using the universal $\mathrm{SL}_2(\mathbb{R})$ deficit identity $\mathrm{Sym}^2(S) - \mathrm{vec}(S)\,\mathrm{vec}(S)^T = \det(S)K_0$ and the Lorentzian Reverse Cauchy–Schwarz Inequality in signature $(+,+,-)$, we prove that for every nonnegative symmetric $n\times n$ matrix G of rank ≤ 2 (both positive semidefinite and indefinite), $\widetilde{\lambda}_2 + \widetilde{\lambda}_{n^2} \geq 0$, resolving Rossman's Conjecture 3.2, Conjec-

ture (∗∗), and Sidorenko's inequality for $K_{5,5} \setminus C_{10}$ and all ML_{2m} unconditionally in rank ≤ 2, as well as Conjecture 4.2 for $n = 2$ across all $t \in [0, 1]$.

(4) **Pointwise Eigenvalue Dominance for All Rank-≤ 3 Positive Semidefinite Host Matrices of Arbitrary Size $n \times n$ (Theorem 5.2):** Using the SO(3)-equivariance of the Hodge star $\star : \wedge^2(\mathbb{R}^3) \xrightarrow{\sim} \mathbb{R}^3$, we prove that in the eigenbasis $\{u_1, u_2, u_3\}$ of $\mathcal{M}_3^- = \sum_{e=1}^m \wedge^2(S_e) = \operatorname{diag}(\tau_3, \tau_2, \tau_1)$ $(\tau_1 \geq \tau_2 \geq \tau_3)$, the compression of $\mathcal{M}_3^+ = \sum_{e=1}^m \operatorname{Sym}^2(S_e)$ to the 4-dimensional traceless subspace $U_4 = \{V \in \operatorname{Sym}_3(\mathbb{R}) : \operatorname{Tr}(V) = 0,\ u_1^T V u_1 = 0\}$ satisfies the exact identity

$$\mathcal{M}_3^+\big|_{U_4} \;=\; \operatorname{diag}(\tau_1,\, \tau_1,\, \tau_2,\, \tau_3) \;+\; \sum_{e=1}^{m} \mathbf{z}_e \mathbf{z}_e^T \;\succeq\; \operatorname{diag}(\tau_1,\, \tau_1,\, \tau_2,\, \tau_3).$$

By Cauchy interlacing and Weyl monotonicity, this proves *pointwise eigenvalue dominance* $\lambda_{k+1}^+(\widetilde{G}) \geq \lambda_k^-(\widetilde{G})$ $(k = 1, 2, 3)$, Rossman's Majorization Conjecture (∗∗), Conjecture 3.2, and Sidorenko's inequality for $K_{5,5} \setminus C_{10}$ and all ML_{2m} for *every* nonnegative symmetric positive semidefinite matrix $G \succeq 0$ of rank ≤ 3 and arbitrary dimension $n \times n$.

(5) **Unconditional Dominance for Zero-Diagonal 3×3 Indefinite Matrices Across All $t \in [0, 1]$ (Theorem 5.4):** For every zero-diagonal 3×3 nonnegative symmetric matrix A (rank 3, indefinite signature $(1, 2)$) and every $t \in [0, 1]$, we prove $\lambda_2^+(M(t)) \geq \lambda_1^-(M(t))$ and establish Conjectures 3.2, 4.2, and (∗∗).

(6) **Analytic Disproof of Rossman's General Conjecture 4.2 for $n \geq 3$ and $t \in (2/3, 1)$ (Theorem 6.1):** Using the exact determinant identity $\det(L_{124} + c^2 I) = 2c^2(c^2(d^2 + c^2) - r^2)$, we give an explicit analytic proof that for every $t \in (2/3, 1)$ there exist strictly positive 3×3 symmetric matrices A such that $|\lambda_9(M(t))| > \lambda_2(M(t))$ and $f_t(p)$ is strictly increasing for large p (and already at $p = 5$ for $t = 0.8$), while showing why the geometric progression $r^2 = sb^2$ at $t = 1/3$ protects $\widetilde{G}$.

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1 Introduction and Problem Formulation

Let $H = (V(H), E(H))$ be a bipartite graph and let $G \in \mathbb{R}_{\geq 0}^{n\times n}$ be a nonnegative symmetric matrix (representing an edge-weighted graph on n vertices, or a discretized symmetric graphon $W : [0,1]^2 \to \mathbb{R}_{\geq 0}$). The unnormalized homomorphism count of H in G is

$$\hom(H, G) \;=\; \sum_{\varphi: V(H)\to\{1,\dots,n\}} \;\prod_{\{u,v\}\in E(H)} G_{\varphi(u),\varphi(v)}.$$

Sidorenko's Conjecture [6] (independently posed in equivalent form by Simonovits [7]) states that for every bipartite graph H and every nonnegative symmetric $n \times n$ matrix G,

$$\frac{\hom(H, G)}{n^{|V(H)|}} \;\geq\; \left(\frac{\hom(K_2, G)}{n^2}\right)^{|E(H)|}. \tag{1}$$

Over the past three decades, Sidorenko's inequality (1) has been established for trees, even cycles, complete bipartite graphs, hypercubes [3], bipartite graphs with a vertex complete to the other part [1], and reflection/tree-decomposable families [8, 2]. While the general conjecture for arbitrary bipartite graphs was recently disproved via a 35-vertex bipartite counterexample, determining which structured families of bipartite graphs satisfy (1) remains a central problem in extremal combinatorics. In particular, the 3-regular 10-vertex graph

$$K_{5,5} \setminus C_{10} \;\cong\; ML_{10},$$

obtained by deleting a Hamiltonian 10-cycle from the complete bipartite graph $K_{5,5}$ (equivalently, the 5-rung *Möbius ladder* ML_{10}), has stood since 1991 as the smallest and most famous individual graph for which Sidorenko's inequality (1) remains open.

1.1 Rossman's Twisted Tensor Operator $\widetilde{G}$ and the Family $M(t)$

In a recent landmark paper, Rossman [4] discovered an exact spectral representation for homomorphism counts of all m-rung bipartite ladders BL_{2m} (where BL_{2m} is the prism graph $C_{2m}\square K_2$ for even m and the Möbius ladder ML_{2m} for odd m, with $BL_{10} = ML_{10} \cong K_{5,5} \setminus C_{10}$ at $m = 5$).

Given a nonnegative symmetric $n \times n$ matrix G, Rossman defines the $n^2 \times n^2$ symmetric matrix $\widetilde{G}$ indexed by pairs $(i, j), (k, l) \in \{1, \dots, n\}^2$:

$$\widetilde{G}_{ij,kl} \;:=\; G_{il}G_{kj}\sqrt{G_{ij}G_{kl}}, \qquad \widetilde{G} \;=\; D(G \otimes G)PD, \tag{2}$$

where $D = \operatorname{diag}(\sqrt{G_{ij}})_{i,j=1}^{n}$ is the $n^2 \times n^2$ diagonal matrix of entrywise square roots and P is the $n^2 \times n^2$ transpose permutation matrix defined by $P_{ij,kl} = \delta_{i,l}\delta_{j,k}$ (so $P\operatorname{vec}(V) = \operatorname{vec}(V^T)$ for all $V \in \mathbb{R}^{n\times n}$). Rossman proved [4, Lemma 2.1] that for every integer $m \geq 1$,

$$\hom(BL_{2m}, G) \;=\; \operatorname{Tr}\left(\widetilde{G}^m\right), \qquad \operatorname{Tr}(\widetilde{G}) \;=\; \sum_{i,j=1}^{n} G_{ij}^3, \qquad \hom(K_{3,3}, G) \;=\; \operatorname{Tr}\left(\widetilde{G}^3\right). \tag{3}$$

Writing $\widetilde{\lambda}_1 \geq \widetilde{\lambda}_2 \geq \cdots \geq \widetilde{\lambda}_{n^2}$ for the real eigenvalues of $\widetilde{G}$ and using Sidorenko's inequality for $K_{3,3} \cong BL_6$ ($\hom(K_{3,3}, G)/n^6 \geq (\hom(K_2, G)/n^2)^9$), Sidorenko's inequality for BL_{2m} ($m \geq 3$) follows from the purely spectral moment inequality

$$\operatorname{Tr}\left(\widetilde{G}^m\right) \;=\; \sum_{k=1}^{n^2} \widetilde{\lambda}_k^m \;\geq\; \left(\operatorname{Tr}(\widetilde{G}^3)\right)^{m/3} \;\geq\; n^{2m}\left(\frac{\hom(K_2, G)}{n^2}\right)^{3m}. \tag{4}$$

For even m, $\widetilde{\lambda}_k^m = |\widetilde{\lambda}_k|^m \geq 0$, so $\operatorname{Tr}(\widetilde{G}^m) \geq 0$ is immediate. For odd $m \geq 3$ (and in particular $m = 5$, corresponding to $K_{5,5} \setminus C_{10}$), the negative eigenvalues of $\widetilde{G}$ enter $\operatorname{Tr}(\widetilde{G}^m)$ with negative

signs. To control these negative eigenvalues, Rossman introduced the concept of a *pseudo-positive spectrum* [4, Definition 3.1]: a real symmetric matrix with eigenvalues $\mu_1, \dots, \mu_N$ has a pseudo-positive spectrum if the signed ℓ_p-norm functional

$$f(p) \;:=\; \left(\sum_{k=1}^{N} \operatorname{sign}(\mu_k) |\mu_k|^p \right)^{1/p} \tag{5}$$

is well-defined, nonnegative, and nonincreasing over $p \in [1, \infty)$. Since $f(5) \le f(3)$ implies $\operatorname{Tr}(\widetilde{G}^5) \ge (\operatorname{Tr}(\widetilde{G}^3))^{5/3} = \hom(K_{3,3}, G)^{5/3} \ge n^{10}(\hom(K_2, G)/n^2)^{15}$, pseudo-positivity of $\widetilde{G}$ immediately implies Sidorenko's inequality for $K_{5,5} \setminus C_{10}$ and all ML_{2m}.

Furthermore, by setting $A := G^{\circ 3/2}$ (so $G = A^{\circ 2/3}$), Rossman embedded $\widetilde{G} = M(1/3)$ into the continuous one-parameter family of $n^2 \times n^2$ matrices $M(t)$ for $t \in [0, 1]$:

$$M(t)_{ij,kl} \;:=\; (A_{ij}A_{kl})^t (A_{il}A_{kj})^{1-t}, \tag{6}$$

interpolating between $M(0) = (A \otimes A)P$ and the rank-1 positive semidefinite matrix $M(1) = \operatorname{vec}(A) \operatorname{vec}(A)^T$. Rossman formulated three central conjectures:

- **Conjecture 3.2 [4]:** For every nonnegative symmetric $n \times n$ matrix G, $\widetilde{G} = M(1/3)$ has a pseudo-positive spectrum.
- **Conjecture 4.2 [4, Abstract & §4]:** For every positive symmetric $n \times n$ matrix A and *every* $t \in [0, 1]$, $M(t)$ has a pseudo-positive spectrum.
- **Majorization Conjecture** $(**)$ **[4, §6, p. 9]:** For every nonnegative symmetric $n \times n$ matrix G and every $k \in \{2, \dots, n^2\}$,

$$(\widetilde{\lambda}_2 + \cdots + \widetilde{\lambda}_k) + (\widetilde{\lambda}_{n^2-k+2} + \cdots + \widetilde{\lambda}_{n^2}) \;\ge\; 0. \tag{7}$$

By Karamata's majorization inequality applied to the convex function $x \mapsto x^p$ ($p \ge 1$), $(**)$ implies $\sum_{k=2}^{n^2} \operatorname{sign}(\widetilde{\lambda}_k) |\widetilde{\lambda}_k|^p \ge 0$, and in fact $(**)$ for $k = 2$ alone ($\widetilde{\lambda}_2 + \widetilde{\lambda}_{n^2} \ge 0$) is a necessary condition for $f(p)$ to be nonincreasing as $p \to \infty$ whenever $\widetilde{\lambda}_2$ and $\widetilde{\lambda}_{n^2}$ are nondegenerate.

1.2 Summary of Main Results

In this work, we develop an exact representation-theoretic, exterior-algebraic, and Lorentzian-geometric framework for $M(t)$ and $\widetilde{G}$ that yields six main theorems:

Theorem 1. (Canonical Symmetric–Exterior Decomposition, Universal Trace Identity, & Extended Star-Subspace Decomposition; Section 2) For all $n \ge 1$ and $t \in [0, 1]$, $M(t) = \mathcal{L}_+(t) \oplus (-\mathcal{L}_-(t))$ on $\mathbb{R}^{n \times n} = \operatorname{Sym}_n(\mathbb{R}) \oplus^{\perp} \operatorname{Skew}_n(\mathbb{R})$, with $|(\mathcal{L}_-(t))_{\alpha,\beta}| \le (\mathcal{L}_+^{\mathrm{off}}(t))_{\alpha,\beta}$ and

$$\operatorname{Tr}\left(\mathcal{L}_+(t) - ww^T\right) \;=\; \operatorname{Tr}\left(\mathcal{L}_-(t)\right) \;=\; \sum_{1 \le i < j \le n} A_{ij}^{2t}(B_{ii}B_{jj} - B_{ij}^2), \qquad B = A^{\circ(1-t)}.$$

Furthermore, $\langle V, (\operatorname{Sym}^2 B - \operatorname{vec}(B) \operatorname{vec}(B)^T)V \rangle \;=\; -2 \operatorname{Tr}((\wedge^2 V)(\wedge^2 B))$ for all $V \in \operatorname{Sym}_n(\mathbb{R})$, and on the n-dimensional extended star subspace $\widehat{\mathcal{S}}_v^+$ of any vertex $v \in \{1, \dots, n\}$, $(\mathcal{L}_+(t) - ww^T)|_{\widehat{\mathcal{S}}_v^+} = 0_1 \oplus \mathcal{L}_-(t)|_{\mathcal{S}_v^-}$.

Theorem 2. (Square-Root-Free Cubic Moment Tensor Reduction; Section 3) By cyclic similarity $\operatorname{Spec}(\widetilde{G}) = \operatorname{Spec}(D^2(G \otimes G)P)$, all square roots in $\widetilde{G}$ disappear from its spectrum. For any rank-r symmetric matrix $G = \sum_{e=1}^{r} \varepsilon_e x^{(e)} (x^{(e)})^T$ ($\varepsilon_e \in \{\pm 1\}$) of arbitrary size $n \times n$, the nonzero spectrum of $\widetilde{G}$ equals the spectrum of the $r^2 \times r^2$ cubic moment operator

$$\mathcal{M}_r \;=\; (J \otimes J)P \sum_{e=1}^{r} \varepsilon_e (S_e \otimes S_e) \;=\; \mathcal{M}_r^+ \;\oplus\; (-\mathcal{N}_r^-),$$

where $(S_e)_{ab} = \sum_{i=1}^{n} x_i^{(e)} x_i^{(a)} x_i^{(b)}$ $(1 \le a, b, e \le r)$ are r symmetric $r \times r$ cubic moment matrices.

Theorem 3. (Unconditional Proof for All Rank-≤ 2 Matrices and Graphons; Section 4) For every nonnegative symmetric $n \times n$ matrix G of rank ≤ 2 (both positive semidefinite and indefinite, for arbitrary $n \ge 2$) and every rank-≤ 2 symmetric graphon W, Rossman's Conjecture 3.2, the Majorization Conjecture $(**)$, and Sidorenko's inequality for $K_{5,5} \setminus C_{10}$ and all bipartite Möbius ladders ML_{2m} hold unconditionally, and Conjecture 4.2 holds for $n = 2$ across all $t \in [0, 1]$.

Theorem 4. (Pointwise Eigenvalue Dominance for All Rank-≤ 3 Positive Semidefinite Host Matrices of Arbitrary Size $n \times n$; Section 5) For any symmetric matrices $S_1, \ldots, S_m \in \mathrm{Sym}_3(\mathbb{R})$, in the Hodge eigenbasis $\{u_1, u_2, u_3\}$ of $\mathcal{M}_3^- = \sum_{e=1}^{m} \wedge^2(S_e) = \mathrm{diag}(\tau_3, \tau_2, \tau_1)$ $(\tau_1 \ge \tau_2 \ge \tau_3)$, the 4×4 compression of $\mathcal{M}_3^+ = \sum_{e=1}^{m} \mathrm{Sym}^2(S_e)$ to the traceless subspace $U_4 = \{V \in \mathrm{Sym}_3(\mathbb{R}) : \mathrm{Tr}(V) = 0,\ u_1^T V u_1 = 0\}$ satisfies the exact identity

$$\mathcal{M}_3^+|_{U_4} \;=\; \mathrm{diag}(\tau_1,\, \tau_1,\, \tau_2,\, \tau_3) \;+\; \sum_{e=1}^{m} \mathbf{z}_e \mathbf{z}_e^T.$$

Combined with Theorem 3.1, this proves pointwise eigenvalue dominance $\lambda_{k+1}^+(\widetilde{G}) \ge \lambda_k^-(\widetilde{G})$ $(k = 1, 2, 3)$, Rossman's Majorization Conjecture $(**)$, Conjecture 3.2, and Sidorenko's inequality for $K_{5,5} \setminus C_{10}$ and all ML_{2m} for *all* nonnegative symmetric positive semidefinite matrices $G \succeq 0$ of rank ≤ 3 and arbitrary dimension $n \times n$.

Theorem 5. (Unconditional Dominance for Zero-Diagonal 3×3 Indefinite Matrices Across All $t \in [0, 1]$; Section 5) For every zero-diagonal 3×3 nonnegative symmetric matrix A (rank 3, indefinite signature $(1, 2)$) and every $t \in [0, 1]$, $\lambda_2^+(M(t)) \ge \lambda_1^-(M(t))$, establishing Conjectures 3.2, 4.2, and $(**)$.

Theorem 6. (Analytic Disproof of Rossman's Conjecture 4.2 for $n \ge 3$ and $t \in (2/3, 1)$; Section 6) For every $n \ge 3$ and every $t \in (2/3, 1)$, there exist strictly positive symmetric $n \times n$ matrices A such that $|\lambda_{n^2}(M(t))| > \lambda_2(M(t)) = \max_{2 \le k \le n^2-1} |\lambda_k(M(t))|$, causing $f_t(p)$ to be strictly increasing for large p (and already at $p = 5$ for $t = 0.8$). (Counterexamples to Conjecture 4.2 for $t > 2/3$ were also independently found numerically by Rossman [5]; Theorem 6.1 provides the exact analytic proof and explains why the geometric progression $r^2 = sb^2$ at $t = 1/3$ prevents this instability for $\widetilde{G}$.)

2 Canonical Symmetric–Exterior Decomposition and Universal Minor Identity

We begin by establishing the canonical block-diagonalization of the entire family $M(t)$ $(t \in [0, 1])$ and the exact relationship between its symmetric and skew-symmetric components via 2×2 minors.

Theorem 2.1 (Canonical Symmetric–Exterior Decomposition and Universal Minor/Trace Identity)**.** *Let $A \in \mathbb{R}_{\ge 0}^{n \times n}$ be any nonnegative symmetric matrix, let $t \in [0, 1]$, and set $B := A^{\circ(1-t)}$ and $D_t := \mathrm{diag}(A_{ij}^t)_{i,j=1}^n$, so that*

$$M(t) \;=\; D_t (B \otimes B) P D_t.$$

Let $\mathbb{R}^{n\times n} = \mathrm{Sym}_n(\mathbb{R}) \oplus^{\perp} \mathrm{Skew}_n(\mathbb{R})$ be the orthogonal decomposition into the eigenspaces of the transpose involution P ($P = +I$ on $\mathrm{Sym}_n(\mathbb{R})$ of dimension $\binom{n+1}{2}$, and $P = -I$ on $\mathrm{Skew}_n(\mathbb{R})$ of dimension $\binom{n}{2}$), equipped with the standard orthonormal bases

$$\mathcal{B}_+ \;=\; \{E_{ii} : 1 \le i \le n\} \;\cup\; \left\{E_{ij}^{+} := \frac{E_{ij}+E_{ji}}{\sqrt{2}} : 1 \le i < j \le n\right\},$$

$$\mathcal{B}_- \;=\; \left\{E_{ij}^{-} := \frac{E_{ij}-E_{ji}}{\sqrt{2}} : 1 \le i < j \le n\right\}.$$

Then:

(a) ***Orthogonal Block-Diagonalization:*** *$M(t)$ commutes with P and decomposes orthogonally as*

$$M(t) \;=\; \mathcal{L}_+(t) \;\oplus\; \big(-\mathcal{L}_-(t)\big), \tag{8}$$

where $\mathcal{L}_+(t) = D_{t,+}(\mathrm{Sym}^2 B)D_{t,+}$ on $\mathrm{Sym}_n(\mathbb{R})$ and $\mathcal{L}_-(t) = D_{t,-}(\wedge^2 B)D_{t,-}$ on $\mathrm{Skew}_n(\mathbb{R})$. Explicitly, for $1 \le i < j \le n$ and $1 \le k < l \le n$,

$$\mathcal{L}_+(t)_{(i,i),(k,k)} \;=\; A_{ii}^t A_{kk}^t B_{ik}^2, \tag{9}$$

$$\mathcal{L}_+(t)_{(i,i),(k,l)} \;=\; \sqrt{2}\, A_{ii}^t A_{kl}^t B_{ik} B_{il}, \tag{10}$$

$$\mathcal{L}_+(t)_{(i,j),(k,l)} \;=\; A_{ij}^t A_{kl}^t (B_{ik}B_{jl} + B_{il}B_{jk}), \tag{11}$$

$$\mathcal{L}_-(t)_{(i,j),(k,l)} \;=\; A_{ij}^t A_{kl}^t (B_{ik}B_{jl} - B_{il}B_{jk}). \tag{12}$$

(b) ***Off-Diagonal Duality and Entrywise Dominance:*** *Let $U = \{(i,j) : 1 \le i < j \le n\}$ and let $\mathcal{L}_+^{\mathrm{off}}(t)$ denote the $\binom{n}{2} \times \binom{n}{2}$ principal submatrix of $\mathcal{L}_+(t)$ indexed by U. Defining $(\mathcal{L}_U)_{(i,j),(k,l)} := A_{ij}^t A_{kl}^t B_{ik}B_{jl}$ and $(M_U)_{(i,j),(k,l)} := A_{ij}^t A_{kl}^t B_{il}B_{jk}$, we have*

$$\mathcal{L}_+^{\mathrm{off}}(t) \;=\; \mathcal{L}_U + M_U, \qquad \mathcal{L}_-(t) \;=\; \mathcal{L}_U - M_U, \tag{13}$$

and since $\mathcal{L}_U, M_U \ge 0$ entrywise,

$$\big|(\mathcal{L}_-(t))_{\alpha,\beta}\big| \;\le\; (\mathcal{L}_+^{\mathrm{off}}(t))_{\alpha,\beta} \qquad \forall \alpha,\beta \in U. \tag{14}$$

(c) ***Rank-1 Deficit Minor Representation and Universal Trace Identity:*** *Since A is symmetric, $w := \mathrm{vec}(A) \in \mathrm{Sym}_n(\mathbb{R})$ has coordinates $w_{(i,i)} = A_{ii} = A_{ii}^t B_{ii}$ and $w_{(i,j)} = \sqrt{2}A_{ij} = \sqrt{2}A_{ij}^t B_{ij}$ in the basis $\mathcal{B}_+$, and $M(1) = ww^T \oplus 0$. Subtracting ww^T from $\mathcal{L}_+(t)$ expresses every entry of $\mathcal{L}_+(t) - ww^T$ purely in terms of 2×2 minors $\Delta_B(i,j;k,l) := B_{ik}B_{jl} - B_{il}B_{jk}$ of B:*

$$\big(\mathcal{L}_+(t) - ww^T\big)_{(i,i),(k,k)} \;=\; A_{ii}^t A_{kk}^t (B_{ik}^2 - B_{ii}B_{kk}) \;=\; -A_{ii}^t A_{kk}^t \Delta_B(i,k;i,k), \tag{15}$$

$$\big(\mathcal{L}_+(t) - ww^T\big)_{(i,i),(k,l)} \;=\; \sqrt{2}\, A_{ii}^t A_{kl}^t (B_{ik}B_{il} - B_{ii}B_{kl}) \;=\; -\sqrt{2}\, A_{ii}^t A_{kl}^t \Delta_B(i,k;l,i), \tag{16}$$

$$\big(\mathcal{L}_+(t) - ww^T\big)_{(i,j),(k,l)} \;=\; A_{ij}^t A_{kl}^t (B_{ik}B_{jl} + B_{il}B_{jk} - 2B_{ij}B_{kl}). \tag{17}$$

In particular, the first n diagonal entries of $\mathcal{L}_+(t) - ww^T$ vanish identically:

$$\big(\mathcal{L}_+(t) - ww^T\big)_{(i,i),(i,i)} \;=\; A_{ii}^{2t}(B_{ii}^2 - B_{ii}^2) \;=\; 0 \qquad (1 \le i \le n),$$

and the remaining $\binom{n}{2}$ diagonal entries on U are identical to the diagonal entries of $\mathcal{L}_-(t)$:

$$\big(\mathcal{L}_+(t) - ww^T\big)_{(i,j),(i,j)} \;=\; A_{ij}^{2t}(B_{ii}B_{jj} - B_{ij}^2) \;=\; \mathcal{L}_-(t)_{(i,j),(i,j)} \qquad (1 \le i < j \le n).$$

Consequently, the universal trace identity holds for all $n \ge 1$ and all $t \in [0,1]$:

$$\boxed{\mathrm{Tr}\,\big(\mathcal{L}_+(t) - ww^T\big) \;=\; \mathrm{Tr}\,\big(\mathcal{L}_-(t)\big) \;=\; \sum_{1\le i<j\le n} A_{ij}^{2t}(B_{ii}B_{jj} - B_{ij}^2).} \tag{18}$$

Proof. (a) Since A is symmetric, $D_t = \operatorname{diag}(A_{ij}^t)$ satisfies $PD_tP = D_t$, and $B = A^{\circ(1-t)}$ is symmetric so $P(B \otimes B)P = B \otimes B$. Thus

$$PM(t) \;=\; PD_t(B \otimes B)PD_t \;=\; D_tP(B \otimes B)PD_t \;=\; D_t(B \otimes B)D_t \;=\; M(t)P.$$

Since $M(t)$ and P commute and $P^2 = I$, $M(t)$ leaves the orthogonal eigenspaces $\operatorname{Sym}_n(\mathbb{R}) = \ker(P - I)$ and $\operatorname{Skew}_n(\mathbb{R}) = \ker(P + I)$ invariant. On $\operatorname{Sym}_n(\mathbb{R})$, $P = +I$, so $M(t)|_{\operatorname{Sym}_n(\mathbb{R})} = \Pi_+ D_t(B \otimes B)D_t\Pi_+ =: \mathcal{L}_+(t)$. On $\operatorname{Skew}_n(\mathbb{R})$, $P = -I$, so $M(t)|_{\operatorname{Skew}_n(\mathbb{R})} = -\Pi_- D_t(B \otimes B)D_t\Pi_- =: -\mathcal{L}_-(t)$. Evaluating the quadratic forms of $D_t(B \otimes B)D_t$ on the orthonormal bases $\mathcal{B}_+$ and $\mathcal{B}_-$ directly yields (9)–(12).

(b) Comparing (11) and (12) on $U = \{(i,j) : 1 \le i < j \le n\}$ gives (13), and since $A_{ij}^t A_{kl}^t B_{ik}B_{jl} \ge 0$ and $A_{ij}^t A_{kl}^t B_{il}B_{jk} \ge 0$, the triangle inequality $|x + y| = x + y \ge |x - y|$ for $x, y \ge 0$ yields (14).

(c) In the basis $\mathcal{B}_+$, $w = \operatorname{vec}(A)$ has inner products $\langle E_{ii}, A\rangle = A_{ii} = A_{ii}^t B_{ii}$ and $\langle E_{ij}^+, A\rangle = \frac{1}{\sqrt{2}}(A_{ij} + A_{ji}) = \sqrt{2}A_{ij} = \sqrt{2}A_{ij}^t B_{ij}$. Subtracting $w_\alpha w_\beta$ from (9)–(11) gives (15)–(17). Setting $k = i$ in (15) gives $A_{ii}^{2t}(B_{ii}^2 - B_{ii}^2) = 0$. Setting $(k,l) = (i,j)$ $(i < j)$ in (17) gives

$$A_{ij}^{2t}(B_{ii}B_{jj} + B_{ij}^2 - 2B_{ij}^2) \;=\; A_{ij}^{2t}(B_{ii}B_{jj} - B_{ij}^2) \;=\; \mathcal{L}_-(t)_{(i,j),(i,j)}.$$

Summing over all $\binom{n+1}{2}$ diagonal entries of $\mathcal{L}_+(t) - ww^T$ establishes (18). □

Corollary 2.2 (Positive Semidefinite Case)**.** *Whenever $B = A^{\circ(1-t)} \succeq 0$ (in particular, at $t = 1/3$, whenever $G = A^{\circ 2/3} \succeq 0$ is positive semidefinite):*

(i) Both $\mathcal{L}_+(t) \succeq 0$ and $\mathcal{L}_-(t) \succeq 0$ are positive semidefinite, so the positive eigenvalues of $M(t)$ are precisely the eigenvalues $\sigma_1 \ge \cdots \ge \sigma_{\binom{n+1}{2}} \ge 0$ of $\mathcal{L}_+(t)$ and the negative eigenvalues of $M(t)$ are precisely $-\tau_1 \le \cdots \le -\tau_{\binom{n}{2}} \le 0$, where τ_k are the eigenvalues of $\mathcal{L}_-(t)$.

(ii) By (14), the spectral radii satisfy $\tau_1 = \rho(\mathcal{L}_-(t)) \le \rho(\mathcal{L}_+^{\mathrm{off}}(t)) \le \rho(\mathcal{L}_+(t)) = \sigma_1$.

(iii) By (18), $\sum_{k=1}^{\binom{n+1}{2}} \sigma_k - \|A\|_F^2 = \sum_{k=1}^{\binom{n}{2}} \tau_k \ge 0$.

Theorem 2.3 (Universal Exterior-Trace Identity and Extended Star-Subspace Decomposition)**.** *Let $A \in \mathbb{R}^{n\times n}_{\ge 0}$ be any nonnegative symmetric matrix, $t \in [0,1]$, $B = A^{\circ(1-t)}$, and $Z := \wedge^2 B \in \operatorname{Sym}_{\binom{n}{2}}(\mathbb{R})$. Then:*

(a) ***Universal Exterior-Trace Identity:*** *For every symmetric matrix $V \in \operatorname{Sym}_n(\mathbb{R})$,*

$$\big\langle V, \big(\operatorname{Sym}^2(B) - \operatorname{vec}(B)\operatorname{vec}(B)^T\big)V\big\rangle \;=\; \operatorname{Tr}(VBVB) - \big(\operatorname{Tr}(VB)\big)^2 \;=\; -2\operatorname{Tr}\Big((\wedge^2 V)(\wedge^2 B)\Big). \tag{19}$$

Consequently, setting $\widetilde{V} := A^{\circ t} \odot V \in \operatorname{Sym}_n(\mathbb{R})$ and $\widetilde{W} := A^{\circ t} \odot W \in \operatorname{Skew}_n(\mathbb{R})$, both $\mathcal{L}_+(t) - ww^T$ and $\mathcal{L}_-(t)$ are governed by the same minor matrix $Z = \wedge^2 B$:

$$\big\langle V, (\mathcal{L}_+(t) - ww^T)V\big\rangle \;=\; -2\operatorname{Tr}\big((\wedge^2\widetilde{V})\,Z\big), \qquad \big\langle W, \mathcal{L}_-(t)W\big\rangle \;=\; \big\langle \widetilde{W}, Z\widetilde{W}\big\rangle.$$

(b) ***Universal Extended Star-Subspace Decomposition:*** *For any vertex $v \in \{1, \ldots, n\}$, define the n-dimensional extended star subspace $\widehat{\mathcal{S}}_v^+ := \mathbb{R}E_{vv} \oplus \mathcal{S}_v^+ \subset \operatorname{Sym}_n(\mathbb{R})$, where $\mathcal{S}_v^+ = \operatorname{span}\{E_{vi}^+ : i \ne v\}$, and the $(n-1)$-dimensional skew star subspace $\mathcal{S}_v^- := \operatorname{span}\{\widetilde{E}_{vi}^- : i \ne v\} \subset \operatorname{Skew}_n(\mathbb{R})$ with $\widetilde{E}_{vi}^- := \frac{1}{\sqrt{2}}(E_{vi} - E_{iv})$. Under the canonical identification $E_{vi}^+ \leftrightarrow \widetilde{E}_{vi}^-$, the $n \times n$ principal submatrix of $\mathcal{L}_+(t) - ww^T$ on $\widehat{\mathcal{S}}_v^+$ is identically*

$$\boxed{\big(\mathcal{L}_+(t) - ww^T\big)\big|_{\widehat{\mathcal{S}}_v^+} \;=\; \begin{pmatrix} 0 & 0 \\ 0 & \mathcal{L}_-(t)|_{\mathcal{S}_v^-} \end{pmatrix}, \qquad \mathcal{L}_+(t)|_{\widehat{\mathcal{S}}_v^+} \;=\; \begin{pmatrix} 0 & 0 \\ 0 & \mathcal{L}_-(t)|_{\mathcal{S}_v^-} \end{pmatrix} + \widehat{w}_v\widehat{w}_v^T,} \tag{20}$$

where $\widehat{w}_v = (A_{vv}, \sqrt{2}A_{v,i_1}, \ldots, \sqrt{2}A_{v,i_{n-1}})^T \in \mathbb{R}^n$.

Proof. (a) Expanding $\mathrm{Tr}(VBVB) - (\mathrm{Tr}(VB))^2 = \sum_{i,j,k,l=1}^{n}(V_{ij}B_{jk}V_{kl}B_{li} - V_{ij}B_{ij}V_{kl}B_{kl})$ and symmetrizing over (i,k) and (j,l) yields

$$-\frac{1}{2}\sum_{i,k,j,l=1}^{n}(V_{ij}V_{kl} - V_{il}V_{jk})(B_{ij}B_{kl} - B_{il}B_{jk}) \;=\; -2\sum_{i<k}\sum_{j<l}(\wedge^2 V)_{(i,k),(j,l)}(\wedge^2 B)_{(i,k),(j,l)},$$

which is (19).

(b) By (15), $(\mathcal{L}_+(t) - ww^T)_{(v,v),(v,v)} = 0$. For any $i \neq v$, setting $(k,l) = (v,i)$ in (16) gives $\Delta_B(v,v;i,v) = B_{vv}B_{vi} - B_{vi}B_{vv} = 0$, so $(\mathcal{L}_+(t) - ww^T)_{(v,v),(v,i)} = 0$. Finally, for any $i,j \in \{1,\dots,n\} \setminus \{v\}$,

$$\begin{aligned}\langle E^+_{vi},\, \mathcal{L}_+(t)E^+_{vj}\rangle - \langle \widetilde{E}^-_{vi},\, \mathcal{L}_-(t)\widetilde{E}^-_{vj}\rangle \;&=\; (A_{vi}A_{vj})^t\Big[(B_{vv}B_{ij} + B_{vj}B_{iv}) - (B_{vv}B_{ij} - B_{vj}B_{iv})\Big]\\ &=\; 2A_{vi}A_{vj} \;=\; w_{vi}w_{vj},\end{aligned}$$

establishing (20). □

3 Square-Root-Free Cubic Moment Tensor Reduction

A major apparent obstacle to analyzing Rossman's matrix $\widetilde{G}_{ij,kl} = G_{il}G_{kj}\sqrt{G_{ij}G_{kl}}$ is the presence of the non-polynomial entrywise square roots $\sqrt{G_{ij}G_{kl}}$. Our second main theorem shows that these square roots are a coordinate artifact of the symmetric gauge $D(G\otimes G)PD$: passing to the similar operator $D^2(G\otimes G)P$ eliminates every square root and reduces $\mathrm{Spec}(\widetilde{G})$ for any rank-r matrix G of arbitrary size $n\times n$ to an $r^2\times r^2$ cubic moment operator.

Theorem 3.1 (Square-Root-Free Cubic Moment Tensor Reduction)**.** *Let $G \in \mathbb{R}^{n\times n}_{\geq 0}$ be a nonnegative symmetric matrix of rank $r \leq n$, with spectral (or signed factor) decomposition*

$$G \;=\; \sum_{e=1}^{r}\varepsilon_e\, x^{(e)}\big(x^{(e)}\big)^T, \qquad x^{(e)} \in \mathbb{R}^n, \quad \varepsilon_e \in \{\pm 1\}, \tag{21}$$

and let $J := \mathrm{diag}(\varepsilon_1,\dots,\varepsilon_r)$. Define the r symmetric $r\times r$ cubic moment matrices $S_1,\dots,S_r \in \mathrm{Sym}_r(\mathbb{R})$ by

$$(S_e)_{ab} \;:=\; \sum_{i=1}^{n} x_i^{(e)}x_i^{(a)}x_i^{(b)} \qquad (1\leq a,b,e\leq r). \tag{22}$$

Then:

(a) ***Exact** $r^2\times r^2$ **Spectral Reduction:** The nonzero eigenvalues of the $n^2\times n^2$ matrix $\widetilde{G} = D(G\otimes G)PD$ (counted with algebraic multiplicity) are identical to the nonzero eigenvalues of the square-root-free $r^2\times r^2$ operator*

$$\mathcal{M}_r \;:=\; (J\otimes J)P_r\sum_{e=1}^{r}\varepsilon_e(S_e\otimes S_e), \tag{23}$$

where P_r is the transpose involution on $\mathbb{R}^{r\times r}$. Explicitly, in coordinates $(a,b),(c,d) \in \{1,\dots,r\}^2$,

$$(\mathcal{M}_r)_{ab,cd} \;=\; \varepsilon_a\varepsilon_b\sum_{e=1}^{r}\varepsilon_e(S_e)_{bc}(S_e)_{ad}.$$

(b) ***Canonical Block Decomposition on** $\mathrm{Sym}_r(\mathbb{R})\oplus\mathrm{Skew}_r(\mathbb{R})$**:** The operator $\mathcal{M}_r$ leaves $\mathrm{Sym}_r(\mathbb{R})$ (dimension $\binom{r+1}{2}$) and $\mathrm{Skew}_r(\mathbb{R})$ (dimension $\binom{r}{2}$) invariant and decomposes as*

$$\mathcal{M}_r \;=\; \mathcal{M}_r^+ \;\oplus\; (-\mathcal{N}_r^-), \tag{24}$$

where

$$\mathcal{M}_r^+ \;=\; \mathrm{Sym}^2(J)\sum_{e=1}^{r}\varepsilon_e\,\mathrm{Sym}^2(S_e), \qquad \mathcal{N}_r^- \;=\; (\wedge^2 J)\sum_{e=1}^{r}\varepsilon_e(\wedge^2 S_e), \tag{25}$$

and $\mathrm{Spec}_{\neq 0}(\mathcal{L}_+) = \mathrm{Spec}_{\neq 0}(\mathcal{M}_r^+)$, $\mathrm{Spec}_{\neq 0}(\mathcal{L}_-) = \mathrm{Spec}_{\neq 0}(\mathcal{N}_r^-)$. *In particular,* $\mathrm{rank}(\mathcal{L}_+) \le \binom{r+1}{2}$ *and* $\mathrm{rank}(\mathcal{L}_-) \le \binom{r}{2}$.

Proof. (a) For any matrices A, B, $\mathrm{Spec}_{\neq 0}(AB) = \mathrm{Spec}_{\neq 0}(BA)$. Applying this to $\widetilde{G} = D\cdot(G\otimes G)PD$ gives $\mathrm{Spec}_{\neq 0}(\widetilde{G}) = \mathrm{Spec}_{\neq 0}(D^2(G\otimes G)P)$, where $D^2 = \mathrm{diag}(G_{ij})_{i,j=1}^n$ contains no square roots. Let $X = [x^{(1)} \mid \cdots \mid x^{(r)}] \in \mathbb{R}^{n\times r}$, so that $G = XJX^T$ and $G\otimes G = (X\otimes X)(J\otimes J)(X^T\otimes X^T)$. Since $(X^T\otimes X^T)P_n = P_r(X^T\otimes X^T)$, we have

$$D^2(G\otimes G)P_n \;=\; D^2(X\otimes X)(J\otimes J)P_r(X^T\otimes X^T).$$

Using $\mathrm{Spec}_{\neq 0}(UV) = \mathrm{Spec}_{\neq 0}(VU)$ with $U = D^2(X\otimes X)\in\mathbb{R}^{n^2\times r^2}$ and $V = (J\otimes J)P_r(X^T\otimes X^T)\in\mathbb{R}^{r^2\times n^2}$, the nonzero spectrum of $\widetilde{G}$ equals that of the $r^2\times r^2$ matrix

$$VU \;=\; (J\otimes J)P_r(X^T\otimes X^T)D^2(X\otimes X).$$

Now expand $D^2 = \mathrm{diag}(G_{ij}) = \sum_{e=1}^r \varepsilon_e(\,\mathrm{diag}(x^{(e)})\otimes\mathrm{diag}(x^{(e)}))$. For each $e\in\{1,\dots,r\}$,

$$(X^T\,\mathrm{diag}(x^{(e)})X)_{ab} \;=\; \sum_{i=1}^{n} x_i^{(a)}x_i^{(e)}x_i^{(b)} \;=\; (S_e)_{ab},$$

and therefore

$$(X^T\otimes X^T)D^2(X\otimes X) \;=\; \sum_{e=1}^{r}\varepsilon_e(X^T\,\mathrm{diag}(x^{(e)})X)\otimes(X^T\,\mathrm{diag}(x^{(e)})X) \;=\; \sum_{e=1}^{r}\varepsilon_e(S_e\otimes S_e).$$

Multiplying on the left by $(J\otimes J)P_r$ gives (23).

(b) Since J and each S_e are symmetric $r\times r$ matrices, $J\otimes J$ and $S_e\otimes S_e$ commute with P_r and preserve $\mathrm{Sym}_r(\mathbb{R})$ and $\mathrm{Skew}_r(\mathbb{R})$. Restricting (23) to $\mathrm{Sym}_r(\mathbb{R})$ where $P_r = +I$ and $\mathrm{Skew}_r(\mathbb{R})$ where $P_r = -I$ yields (24)–(25). □

Remark 3.2 (Continuous Finite-Rank Graphons). Theorem 3.1 extends verbatim to any symmetric graphon $W:[0,1]^2\to\mathbb{R}_{\ge 0}$ of finite rank $W(x,y) = \sum_{e=1}^r \varepsilon_e\phi_e(x)\phi_e(y)$ ($\phi_e\in L^3([0,1])$), with $(S_e)_{ab} = \int_0^1 \phi_e(x)\phi_a(x)\phi_b(x)\,dx$. Thus, for all finite-rank graphons, Sidorenko's Conjecture for BL_{2m} depends *solely* on the cubic moment tensor $T_{eab} = \int_0^1 \phi_e\phi_a\phi_b\,dx$.

4 Unconditional Proof of Sidorenko's Conjecture for $K_{5,5}\setminus C_{10}$ and Rossman's Conjectures in Rank ≤ 2

We now apply Theorems 2.1 and 3.1 to prove Rossman's Conjecture 3.2, the Majorization Conjecture (∗∗), and Sidorenko's Conjecture for $K_{5,5}\setminus C_{10}$ and all bipartite Möbius ladders ML_{2m} unconditionally for all matrices of rank ≤ 2 of arbitrary dimension $n\times n$, as well as Conjecture 4.2 for $n=2$ across all $t\in[0,1]$.

4.1 Warm-Up: Exact Proof of Conjecture 4.2 for $n=2$ Across All $t\in[0,1]$

Proposition 4.1 (Complete Resolution of Conjecture 4.2 for $n=2$)**.** *Let* $A = \left(\begin{smallmatrix} a & b \\ b & d\end{smallmatrix}\right)$ *be any positive symmetric* 2×2 *matrix* $(a,b,d>0)$ *and let* $t\in[0,1]$. *Set* $r := (ad)^t(b^2)^{1-t}$ *and* $s := (b^2)^t(ad)^{1-t}$. *Then the four eigenvalues* $\lambda_1\ge\lambda_2\ge\lambda_3\ge\lambda_4$ *of* $M(t)$ *satisfy*

$$\lambda_2+\lambda_4 \;\ge\; 0 \qquad \textit{and} \qquad \lambda_3 \;\ge\; 0 \quad (\textit{hence } \lambda_2+\lambda_3+\lambda_4\ge 0),$$

and consequently $M(t)$ *satisfies the majorization condition* (∗∗) *and has a pseudo-positive spectrum for every* $t\in[0,1]$.

Proof. By Theorem 2.1, $M(t) = \mathcal{L}_+(t) \oplus (-\mathcal{L}_-(t))$, where $\mathcal{L}_-(t)$ is the 1×1 scalar

$$\mathcal{L}_-(t) \;=\; s - b^2, \qquad -\mathcal{L}_-(t) \;=\; b^2 - s,$$

and $\mathcal{L}_+(t) - ww^T$ (with $w = (a, d, \sqrt{2}b)^T$) is the 3×3 matrix

$$\mathcal{L}_+(t) - ww^T \;=\; \begin{pmatrix} 0 & r - ad & 0 \\ r - ad & 0 & 0 \\ 0 & 0 & s - b^2 \end{pmatrix},$$

whose exact eigenvalues are $+|ad - r|$, $s - b^2$, and $-|ad - r|$. Setting $x = (ad)^t$, $y = (b^2)^t$, $u = (ad)^{1-t}$, $v = (b^2)^{1-t}$, we have the Muirhead factorization

$$(ad - r) - (s - b^2) \;=\; (x - y)(u - v) \;=\; \big((ad)^t - (b^2)^t\big)\big((ad)^{1-t} - (b^2)^{1-t}\big) \;\geq\; 0, \tag{26}$$

since $z \mapsto z^t$ and $z \mapsto z^{1-t}$ are both nondecreasing on $(0, \infty)$.

Case 1 ($ad \geq b^2$): Then $ad \geq r$ and $s \geq b^2$, and by (26), $ad - r \geq s - b^2 \geq 0$. Thus the eigenvalues of $\mathcal{L}_+(t) - ww^T$ in descending order are

$$\mu_1 = ad - r \;\geq\; \mu_2 = s - b^2 \;\geq\; 0 \;\geq\; \mu_3 = -(ad - r).$$

Furthermore, since $B = A^{\circ(1-t)} = \left(\begin{smallmatrix} a^{1-t} & b^{1-t} \\ b^{1-t} & d^{1-t} \end{smallmatrix}\right)$ has $\det(B) = (ad)^{1-t} - (b^2)^{1-t} \geq 0$, $B \succeq 0$ and hence $\mathcal{L}_+(t) \succeq 0$ has three nonnegative eigenvalues $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$, while $-\mathcal{L}_-(t)$ has the single negative eigenvalue $\lambda_4 = -(s - b^2) \leq 0$. By Weyl's Monotonicity Theorem for $\mathcal{L}_+(t) = (\mathcal{L}_+(t) - ww^T) + ww^T$ (since $ww^T \succeq 0$),

$$\lambda_2 \;=\; \sigma_2(\mathcal{L}_+(t)) \;\geq\; \mu_2(\mathcal{L}_+(t) - ww^T) \;=\; s - b^2 \;=\; |\lambda_4|,$$

so $\lambda_2 + \lambda_4 \geq 0$ and $\lambda_3 = \sigma_3 \geq 0$.

Case 2 ($ad < b^2$): Then $r > ad$ and $b^2 > s$, so $-\mathcal{L}_-(t) = b^2 - s > 0$ is *positive*, and by (26), $b^2 - s \geq r - ad > 0$. The eigenvalues of $\mathcal{L}_+(t) - ww^T$ in descending order are

$$\mu_1 = r - ad \;>\; 0 \;>\; \mu_2 = -(r - ad) \;\geq\; \mu_3 = -(b^2 - s).$$

Moreover, since B has signature $(+, -)$, $\mathrm{Sym}^2(B)$ has signature $(+, +, -)$, so $\mathcal{L}_+(t)$ has two positive eigenvalues $\sigma_1 \geq \sigma_2 > 0$ and one negative eigenvalue $\lambda_4 = \sigma_3(\mathcal{L}_+(t)) < 0$, while $-\mathcal{L}_-(t) = b^2 - s > 0$ is positive. Thus $\lambda_2 = \max(\sigma_2, b^2 - s) \geq b^2 - s$ and $\lambda_3 = \min(\sigma_2, b^2 - s) > 0$. Applying Weyl's Monotonicity Theorem to the minimum eigenvalue of $\mathcal{L}_+(t) = (\mathcal{L}_+(t) - ww^T) + ww^T$ yields

$$\lambda_4 \;=\; \sigma_3(\mathcal{L}_+(t)) \;\geq\; \mu_3(\mathcal{L}_+(t) - ww^T) \;=\; -(b^2 - s) \;\geq\; -\lambda_2,$$

so $\lambda_2 + \lambda_4 \geq 0$ and $\lambda_3 > 0$. In both cases, $\lambda_2 \geq |\lambda_4|$ and $\lambda_3 \geq 0$, which implies $(**)$ and pseudo-positivity of $M(t)$ for all $t \in [0, 1]$. □

4.2 General Rank-≤ 2 Matrices of Arbitrary Dimension $n \times n$: The $\mathrm{SL}_2(\mathbb{R})$ Lorentzian Proof

When $n \geq 3$ and $\mathrm{rank}(G) = 2$, G is an $n \times n$ matrix, so $\mathcal{L}_+$ has dimension $\binom{n+1}{2}$ and $\mathcal{L}_-$ has dimension $\binom{n}{2}$. Remarkably, the $n = 2$ phenomenon generalizes to *arbitrary dimension* n via an exact Lorentzian structure on $\mathrm{Sym}_2(\mathbb{R})$.

Lemma 4.2 (Universal $\mathrm{SL}_2(\mathbb{R})$ Rank-1 Deficit Identity)**.** *For any 2×2 symmetric matrix $S = \left(\begin{smallmatrix} s_{11} & s_{12} \\ s_{12} & s_{22} \end{smallmatrix}\right) \in \mathrm{Sym}_2(\mathbb{R})$, let $\mathrm{vec}(S) = (s_{11}, s_{22}, \sqrt{2}s_{12})^T$ be its coordinate vector in the orthonormal*

basis $\{E_{11}, E_{22}, E_{12}^+\}$ *of* $\mathrm{Sym}_2(\mathbb{R})$*, and let* $\mathrm{Sym}^2(S)$ *be the* 3×3 *matrix of* $S \otimes S$ *restricted to* $\mathrm{Sym}_2(\mathbb{R})$*:*

$$\mathrm{Sym}^2(S) \;=\; \begin{pmatrix} s_{11}^2 & s_{12}^2 & \sqrt{2}s_{11}s_{12} \\ s_{12}^2 & s_{22}^2 & \sqrt{2}s_{22}s_{12} \\ \sqrt{2}s_{11}s_{12} & \sqrt{2}s_{22}s_{12} & s_{11}s_{22} + s_{12}^2 \end{pmatrix}.$$

Then we have the universal polynomial identity

$$\boxed{\mathrm{Sym}^2(S) - \mathrm{vec}(S)\,\mathrm{vec}(S)^T \;=\; \det(S)\,K_0, \qquad K_0 \;:=\; \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},} \tag{27}$$

and the Lorentzian quadratic form $\langle u, v\rangle_{K_0} := u^T K_0 v = -u_1v_2 - u_2v_1 + u_3v_3$ *satisfies*

$$\langle \mathrm{vec}(S), \mathrm{vec}(S)\rangle_{K_0} \;=\; -2(s_{11}s_{22} - s_{12}^2) \;=\; -2\det(S). \tag{28}$$

Proof. Subtracting $\mathrm{vec}(S)\,\mathrm{vec}(S)^T$ from $\mathrm{Sym}^2(S)$ cancels entries $(1,1)$ and $(2,2)$ on the diagonal and $(1,3), (2,3)$ off the diagonal, leaving $s_{12}^2 - s_{11}s_{22} = -\det(S)$ in entries $(1,2)$ and $(2,1)$, and $(s_{11}s_{22} + s_{12}^2) - 2s_{12}^2 = \det(S)$ in entry $(3,3)$. Evaluating $\mathrm{vec}(S)^T K_0\,\mathrm{vec}(S) = -2s_{11}s_{22} + (\sqrt{2}s_{12})^2 = -2\det(S)$ gives (28). □

Theorem 4.3 (Unconditional Proof of Conjecture 3.2, $(**)$, and Sidorenko's Conjecture for $K_{5,5} \setminus C_{10}$ in Rank ≤ 2)**.** *Let* $G \in \mathbb{R}^{n\times n}_{\ge 0}$ *be any nonnegative symmetric* $n \times n$ *matrix of rank* $r \le 2$ *(*$n \ge 2$*). Then:*

(a) If $r \le 1$*, then* $\mathcal{L}_- = 0$ *and* $\widetilde{G} = \mathcal{L}_+ \oplus 0 \succeq 0$ *has rank* ≤ 1 *with a single positive eigenvalue* $\widetilde{\lambda}_1 = \mathrm{Tr}(\widetilde{G})$ *and all other eigenvalues* 0*.*

(b) If $r = 2$*, write* $G = x^{(1)}(x^{(1)})^T + \varepsilon x^{(2)}(x^{(2)})^T$ *with* $\varepsilon \in \{+1, -1\}$ *(where* $x^{(1)} \in \mathbb{R}^n_{\ge 0}$ *is the Perron–Frobenius eigenvector block, so* $S_1 \succ 0$ *and* $\det(S_1) > 0$*). Let* $S_1, S_2 \in \mathrm{Sym}_2(\mathbb{R})$ *be the* 2×2 *cubic moment matrices (22), and set*

$$p_{ab} \;:=\; \langle \mathrm{vec}(S_a), \mathrm{vec}(S_b)\rangle_{K_0} \qquad (a, b \in \{1, 2\}).$$

Then:

(i) $\mathcal{L}_-$ *has rank at most* 1*, with single potentially nonzero eigenvalue*

$$\tau_1 \;=\; \varepsilon\big(\det(S_1) + \varepsilon \det(S_2)\big). \tag{29}$$

(ii) $\mathcal{L}_+ - ww^T$ *has rank at most* 3*, with exact nonzero eigenvalues*

$$\mathrm{Spec}_{\neq 0}\left(\mathcal{L}_+ - ww^T\right) \;=\; \{+\alpha,\ \tau_1,\ -\alpha\}, \tag{30}$$

where $\alpha \ge 0$ *satisfies the exact identity*

$$\boxed{\alpha^2 - \tau_1^2 \;=\; \varepsilon(p_{12}^2 - p_{11}p_{22}).} \tag{31}$$

(iii) By the Lorentzian Reverse Cauchy–Schwarz Inequality in signature $(+,+,-)$*, since* $p_{11} = -2\det(S_1) < 0$ *is timelike,*

$$p_{12}^2 - p_{11}p_{22} \;\ge\; 0 \qquad \textit{unconditionally for all } S_1 \succ 0 \textit{ and all } S_2 \in \mathrm{Sym}_2(\mathbb{R}). \tag{32}$$

(iv) Consequently, in both the positive semidefinite case ($\varepsilon = +1$*, where* $\alpha \ge \tau_1 > 0$*) and the indefinite case (*$\varepsilon = -1$*, where* $\alpha \le |\tau_1|$*), Weyl's Monotonicity Theorem yields*

$$\widetilde{\lambda}_2 + \widetilde{\lambda}_{n^2} \;\ge\; 0 \qquad \textit{and} \qquad \widetilde{\lambda}_k \;\ge\; 0 \quad (2 \le k \le n^2 - 1). \tag{33}$$

(c) In particular, for every nonnegative symmetric $n \times n$ matrix G of rank ≤ 2, Rossman's Majorization Conjecture (**), Conjecture 3.2 (pseudo-positivity of $\widetilde{G}$), and Sidorenko's Conjecture for $K_{5,5} \setminus C_{10}$ and all bipartite Möbius ladders ML_{2m} hold unconditionally.

Proof. Part (a) is immediate from Theorem 3.1(b) since $\binom{1}{2} = 0$. For Part (b), let $r = 2$ and $\varepsilon \in \{+1, -1\}$.

Proof of (i): By Theorem 3.1(b), $\mathrm{Spec}_{\neq 0}(\mathcal{L}_-) = \mathrm{Spec}_{\neq 0}(\mathcal{N}_2^-)$ where $\mathcal{N}_2^- = (\wedge^2 J)(\wedge^2 S_1 + \varepsilon \wedge^2 S_2)$ on the 1-dimensional space $\mathrm{Skew}_2(\mathbb{R})$. Since $\wedge^2 J = \det(J) = \varepsilon$ and $\wedge^2 S_e = \det(S_e)$, $\mathcal{N}_2^-$ is the 1×1 scalar $\tau_1 = \varepsilon(\det(S_1) + \varepsilon \det(S_2))$, proving (29).

Proof of (ii): Since $G = XJX^T$ with $J = \mathrm{diag}(1, \varepsilon)$, in $\mathrm{Sym}_n(\mathbb{R})$ we have $\mathcal{L}_+ = \Pi_+ D(X \otimes X)\,\mathrm{Sym}^2(J)(X^T \otimes X^T) D\Pi_+$ and $w = \Pi_+ D(X \otimes X)\,\mathrm{vec}(J)$, so

$$\mathcal{L}_+ - ww^T \;=\; \Pi_+ D(X \otimes X)\Big(\mathrm{Sym}^2(J) - \mathrm{vec}(J)\,\mathrm{vec}(J)^T\Big)(X^T \otimes X^T) D\Pi_+.$$

Applying Lemma 4.2 directly to the 2×2 matrix $J = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \varepsilon \end{smallmatrix}\right)$ (which has $\det(J) = \varepsilon$) gives the exact operator factorization

$$\boxed{\mathrm{Sym}^2(J) - \mathrm{vec}(J)\,\mathrm{vec}(J)^T \;=\; \det(J)\,K_0 \;=\; \varepsilon K_0.} \tag{34}$$

Therefore, by cyclic invariance of the nonzero spectrum $\mathrm{Spec}_{\neq 0}(U(\varepsilon K_0)U^T) = \mathrm{Spec}_{\neq 0}(\varepsilon K_0 U^T U)$ where $U = \Pi_+ D(X \otimes X)$ and $U^T U = \sum_{e=1}^2 \varepsilon_e\,\mathrm{Sym}^2(S_e)$, the nonzero eigenvalues of $\mathcal{L}_+ - ww^T$ are identical to the eigenvalues of the 3×3 matrix

$$\mathcal{R}_3 \;:=\; \varepsilon K_0\Big(\mathrm{Sym}^2(S_1) + \varepsilon\,\mathrm{Sym}^2(S_2)\Big). \tag{35}$$

Now substitute Lemma 4.2 ($\mathrm{Sym}^2(S_e) = \mathrm{vec}(S_e)\,\mathrm{vec}(S_e)^T + \det(S_e)K_0$) into (35) and use $K_0^2 = I_3$:

$$\begin{aligned} \mathcal{R}_3 \;&=\; \varepsilon K_0\Big(\mathrm{vec}(S_1)\,\mathrm{vec}(S_1)^T + \varepsilon\,\mathrm{vec}(S_2)\,\mathrm{vec}(S_2)^T + \big(\det(S_1) + \varepsilon\det(S_2)\big)K_0\Big) \\ &=\; \tau_1 I_3 \;+\; \varepsilon K_0(v_1 v_1^T + \varepsilon v_2 v_2^T), \end{aligned} \tag{36}$$

where $v_e := \mathrm{vec}(S_e) \in \mathbb{R}^3$ and $\tau_1 = \varepsilon(\det(S_1) + \varepsilon\det(S_2))$. Equation (36) shows that $\mathcal{R}_3 - \tau_1 I_3 = \varepsilon K_0(v_1 v_1^T + \varepsilon v_2 v_2^T)$ has rank at most 2 (since $\mathrm{span}(v_1, v_2)$ is at most 2-dimensional), so 0 is an eigenvalue of $\mathcal{R}_3 - \tau_1 I_3$, meaning τ_1 is always an eigenvalue of $\mathcal{R}_3$. On the 2-dimensional invariant subspace $\mathrm{span}(K_0 v_1, K_0 v_2)$, the action of $\mathcal{R}_3 - \tau_1 I_3 = \varepsilon K_0(v_1 v_1^T + \varepsilon v_2 v_2^T)$ has the 2×2 matrix representation

$$Q_2 \;=\; \varepsilon \begin{pmatrix} v_1^T K_0 v_1 & \varepsilon v_1^T K_0 v_2 \\ v_2^T K_0 v_1 & \varepsilon v_2^T K_0 v_2 \end{pmatrix} \;=\; \varepsilon \begin{pmatrix} p_{11} & \varepsilon p_{12} \\ p_{12} & \varepsilon p_{22} \end{pmatrix}.$$

Let us compute the trace and determinant of $\mathcal{R}_3 - \tau_1 I_3$ via Q_2:

- By (28), $p_{11} = -2\det(S_1)$ and $p_{22} = -2\det(S_2)$, so

$$\mathrm{Tr}(Q_2) \;=\; \varepsilon(p_{11} + \varepsilon p_{22}) \;=\; -2\varepsilon(\det(S_1) + \varepsilon\det(S_2)) \;=\; -2\tau_1.$$

- And the determinant of Q_2 is

$$\det(Q_2) \;=\; \varepsilon^2(\varepsilon p_{11} p_{22} - \varepsilon p_{12}^2) \;=\; -\varepsilon(p_{12}^2 - p_{11}p_{22}).$$

Therefore, the two remaining eigenvalues μ_1, μ_2 of $\mathcal{R}_3$ satisfy $(\mu_i - \tau_1)^2 - \mathrm{Tr}(Q_2)(\mu_i - \tau_1) + \det(Q_2) = 0$, i.e.,

$$(\mu_i - \tau_1)^2 + 2\tau_1(\mu_i - \tau_1) - \varepsilon(p_{12}^2 - p_{11}p_{22}) \;=\; 0 \quad\Longleftrightarrow\quad \mu_i^2 \;=\; \tau_1^2 + \varepsilon(p_{12}^2 - p_{11}p_{22}) \;=:\; \alpha^2.$$

Thus the three eigenvalues of $\mathcal{R}_3$ (and hence the nonzero eigenvalues of $\mathcal{L}_+ - ww^T$) are $\{+\alpha, \tau_1, -\alpha\}$ with $\alpha^2 - \tau_1^2 = \varepsilon(p_{12}^2 - p_{11}p_{22})$, proving (30) and (31).

Proof of (iii): In the orthonormal coordinates $t_a = \frac{(S_a)_{11}+(S_a)_{22}}{\sqrt{2}}$, $x_a = \frac{(S_a)_{11}-(S_a)_{22}}{\sqrt{2}}$, $y_a = \sqrt{2}(S_a)_{12}$ ($a \in \{1,2\}$), the form K_0 diagonalizes as $\mathrm{diag}(-1,+1,+1)$, so $p_{ab} = -t_a t_b + x_a x_b + y_a y_b$. By the exact Brahmagupta–Minkowski sum-of-squares identity (formally verified in Lean 4 in `SidorenkoMobiusTheorem.lean`),

$$t_1^2(p_{12}^2 - p_{11}p_{22}) \;=\; (-p_{11})\Big[(t_1x_2 - t_2x_1)^2 + (t_1y_2 - t_2y_1)^2\Big] + \big(x_1(t_1x_2 - t_2x_1) + y_1(t_1y_2 - t_2y_1)\big)^2. \tag{37}$$

Since $S_1 = \sum_{i=1}^n x_i^{(1)} \begin{pmatrix} x_i^{(1)} \\ x_i^{(2)} \end{pmatrix} \begin{pmatrix} x_i^{(1)} & x_i^{(2)} \end{pmatrix} \succ 0$ is a positive-definite Gram matrix, $\det(S_1) > 0$ and $\mathrm{Tr}(S_1) = \sqrt{2}t_1 > 0$, so $-p_{11} = 2\det(S_1) > 0$. Every term on the right-hand side of (37) is a nonnegative square, whence $p_{12}^2 - p_{11}p_{22} \geq 0$.

Proof of (iv) and (c):

- **Case 1 ($\varepsilon = +1$, $G \succeq 0$):** Here $G \succeq 0$, so $\mathcal{L}_+ \succeq 0$ and $\mathcal{L}_- \succeq 0$ has the single positive eigenvalue $\tau_1 = \det(S_1) + \det(S_2) > 0$, making $\widetilde{\lambda}_{n^2} = -\tau_1$ the unique negative eigenvalue of $\widetilde{G}$. By (31) and (32), $\alpha^2 - \tau_1^2 = p_{12}^2 - p_{11}p_{22} \geq 0$, so $\alpha \geq \tau_1 > 0$. Thus the two positive eigenvalues of $\mathcal{L}_+ - ww^T$ are $\mu_1 = \alpha \geq \mu_2 = \tau_1 > 0$. Since $ww^T \succeq 0$, Weyl's Monotonicity Theorem gives
$$\widetilde{\lambda}_2 \;=\; \sigma_2(\mathcal{L}_+) \;\geq\; \mu_2(\mathcal{L}_+ - ww^T) \;=\; \tau_1 \;=\; |\widetilde{\lambda}_{n^2}|,$$
and all other eigenvalues of $\widetilde{G}$ are nonnegative ($\sigma_k(\mathcal{L}_+) \geq 0$).

- **Case 2 ($\varepsilon = -1$, G indefinite of rank 2):** Since $G_{ij} = x_i^{(1)}x_j^{(1)} - x_i^{(2)}x_j^{(2)} \geq 0$, we have $|x_i^{(2)}| \leq x_i^{(1)}$ for all i. By the triangle inequality in the positive semidefinite cone $\mathrm{Sym}_2(\mathbb{R})_{\succeq 0}$,
$$\pm S_2 \;=\; \sum_{i=1}^n (\pm x_i^{(2)}) \begin{pmatrix} x_i^{(1)} \\ x_i^{(2)} \end{pmatrix} \begin{pmatrix} x_i^{(1)} & x_i^{(2)} \end{pmatrix} \;\preceq\; \sum_{i=1}^n x_i^{(1)} \begin{pmatrix} x_i^{(1)} \\ x_i^{(2)} \end{pmatrix} \begin{pmatrix} x_i^{(1)} & x_i^{(2)} \end{pmatrix} \;=\; S_1,$$
so $-S_1 \preceq S_2 \preceq S_1$, which implies $|\det(S_2)| \leq \det(S_1)$ and hence $\det(S_1) - \det(S_2) \geq 0$. Thus $\tau_1 = -(\det(S_1) - \det(S_2)) \leq 0$, so $-\mathcal{L}_-$ has the single *nonnegative* eigenvalue $\lambda_{\mathrm{skew}} := -\tau_1 = \det(S_1) - \det(S_2) \geq 0$ (and no negative eigenvalues!). Meanwhile, by (31) and (32) with $\varepsilon = -1$,
$$\alpha^2 - \tau_1^2 \;=\; -(p_{12}^2 - p_{11}p_{22}) \;\leq\; 0 \quad\Longrightarrow\quad 0 \;\leq\; \alpha \;\leq\; |\tau_1| \;=\; \lambda_{\mathrm{skew}}.$$
Thus the minimum eigenvalue of $\mathcal{L}_+ - ww^T$ is $\mu_{\min}(\mathcal{L}_+ - ww^T) = \min(-\alpha, \tau_1) = \tau_1 = -\lambda_{\mathrm{skew}}$. Moreover, since $\mathrm{Sym}^2(J) = \mathrm{diag}(1,1,-1)$ has signature $(+,+,-)$, Sylvester's Law of Inertia implies $\mathcal{L}_+$ has at most one negative eigenvalue $\widetilde{\lambda}_{n^2} = \sigma_{\min}(\mathcal{L}_+)$. By Weyl's Monotonicity Theorem for $\mathcal{L}_+ = (\mathcal{L}_+ - ww^T) + ww^T$,
$$\widetilde{\lambda}_{n^2} \;=\; \sigma_{\min}(\mathcal{L}_+) \;\geq\; \mu_{\min}(\mathcal{L}_+ - ww^T) \;=\; -\lambda_{\mathrm{skew}} \;\geq\; -\widetilde{\lambda}_2,$$
where the last inequality follows from $\widetilde{\lambda}_2 \geq \lambda_{\mathrm{skew}}$ (since λ_{skew} is an eigenvalue of $\widetilde{G}$ separate from the Perron root $\widetilde{\lambda}_1 = \sigma_1(\mathcal{L}_+)$).

In both cases, $\widetilde{G}$ has at most one negative eigenvalue $\widetilde{\lambda}_{n^2}$ and satisfies $\widetilde{\lambda}_2 \geq |\widetilde{\lambda}_{n^2}|$, establishing $(**)$, Conjecture 3.2, and Sidorenko's Conjecture for $K_{5,5} \setminus C_{10}$ and all ML_{2m}. □

5 Pointwise Eigenvalue Dominance Beyond Rank 2: Rank-$\leq$ 3 Positive Semidefinite Host Matrices and Indefinite 3×3 Families

We now extend our unconditional proofs beyond rank 2. By Theorem 3.1, for any positive semidefinite nonnegative symmetric matrix $G = \sum_{e=1}^{r} x^{(e)}(x^{(e)})^T \succeq 0$ of rank r and arbitrary dimension $n \times n$, the nonzero spectrum of $\widetilde{G}$ coincides with that of the $r^2 \times r^2$ cubic moment operator

$$\mathcal{M}_r \;=\; \mathcal{M}_r^+ \;\oplus\; (-\mathcal{M}_r^-), \qquad \mathcal{M}_r^+ \;=\; \sum_{e=1}^{r} \mathrm{Sym}^2(S_e), \qquad \mathcal{M}_r^- \;=\; \sum_{e=1}^{r} \wedge^2(S_e),$$

where $(S_e)_{ab} = \sum_{i=1}^{n} x_i^{(e)} x_i^{(a)} x_i^{(b)}$ $(1 \leq a, b, e \leq r)$. Moreover, setting $x_i := (x_i^{(1)}, \ldots, x_i^{(r)})^T \in \mathbb{R}^r$, the cubic moment operators admit the direct positive-semidefinite rank-1 decompositions

$$\mathcal{M}_r^+ \;=\; \sum_{i,j=1}^{n} G_{ij}\,(x_i \vee x_j)(x_i \vee x_j)^T \;\succeq\; 0, \qquad \mathcal{M}_r^- \;=\; \sum_{i,j=1}^{n} G_{ij}\,(x_i \wedge x_j)(x_i \wedge x_j)^T \;\succeq\; 0, \tag{38}$$

where $(x_i \vee x_j)_{aa} = x_{i,a}x_{j,a}$, $(x_i \vee x_j)_{ab} = \frac{1}{\sqrt{2}}(x_{i,a}x_{j,b} + x_{i,b}x_{j,a})$ $(a < b)$, and $(x_i \wedge x_j)_{ab} = \frac{1}{\sqrt{2}}(x_{i,a}x_{j,b} - x_{i,b}x_{j,a})$ $(a < b)$. In particular, when $G \succeq 0$, $\mathcal{M}_r^+ \succeq 0$ has no negative eigenvalues, so the negative eigenvalues of $\widetilde{G}$ are precisely $-\lambda_1(\mathcal{M}_r^-) \leq -\lambda_2(\mathcal{M}_r^-) \leq \cdots \leq -\lambda_{\binom{r}{2}}(\mathcal{M}_r^-)$.

5.1 Universal Simple-Bivector 2×2 Traceless Dominance for Arbitrary Rank $r \geq 2$

We first establish a universal rank-r comparison between $\mathcal{M}_r^+$ and $\mathcal{M}_r^-$ along any simple (decomposable) bivector $x \wedge y \in \wedge^2(\mathbb{R}^r)$.

Lemma 5.1 (Universal Simple-Bivector 2×2 Traceless Dominance)**.** *Let $r \geq 2$ and let $S_1, \ldots, S_m \in \mathrm{Sym}_r(\mathbb{R})$ be arbitrary real symmetric $r \times r$ matrices. For any orthonormal pair of vectors $x, y \in \mathbb{R}^r$ ($\|x\| = \|y\| = 1$, $x \perp y$) spanning the 2-plane $\Sigma = \mathrm{span}\{x, y\}$ with unit simple bivector $w_\Sigma = x \wedge y \in \wedge^2(\mathbb{R}^r)$, let*

$$\mathcal{Q}_\Sigma \;:=\; \mathrm{span}\,\{Q_1, Q_2\} \;\subset\; \mathrm{Sym}_r(\mathbb{R}), \qquad Q_1 \;:=\; \frac{xx^T - yy^T}{\sqrt{2}}, \qquad Q_2 \;:=\; \frac{xy^T + yx^T}{\sqrt{2}},$$

be the 2-dimensional orthonormal subspace of traceless symmetric matrices supported on Σ. Then the 2×2 compression of $\mathcal{M}_r^+ = \sum_{e=1}^{m} \mathrm{Sym}^2(S_e)$ to $\mathcal{Q}_\Sigma$ satisfies the exact operator identity

$$\boxed{\mathcal{M}_r^+\big|_{\mathcal{Q}_\Sigma} \;=\; \langle w_\Sigma, \mathcal{M}_r^- w_\Sigma\rangle I_2 \;+\; \sum_{e=1}^{m} u_e u_e^T \;\succeq\; \langle w_\Sigma, \mathcal{M}_r^- w_\Sigma\rangle I_2,} \tag{39}$$

where $u_e := \Big(\frac{x^T S_e x - y^T S_e y}{\sqrt{2}},\, \sqrt{2}\, x^T S_e y\Big)^T \in \mathbb{R}^2$. Consequently, by Cauchy's Interlacing Theorem,

$$\lambda_2(\mathcal{M}_r^+) \;\geq\; \lambda_{\min}\Big(\mathcal{M}_r^+\big|_{\mathcal{Q}_\Sigma}\Big) \;\geq\; \langle x \wedge y, \mathcal{M}_r^-(x \wedge y)\rangle. \tag{40}$$

Proof. For each $e \in \{1, \ldots, m\}$, write the 2×2 restriction of S_e to $\mathrm{span}\{x, y\}$ as $\widehat{S}_e = \left(\begin{smallmatrix} \alpha_e & \beta_e \\ \beta_e & \gamma_e \end{smallmatrix}\right)$, where $\alpha_e = x^T S_e x$, $\beta_e = x^T S_e y$, and $\gamma_e = y^T S_e y$. Then $\langle w_\Sigma, (\wedge^2 S_e) w_\Sigma\rangle = \det(\widehat{S}_e) = \alpha_e\gamma_e - \beta_e^2$.

Meanwhile, for $Q_1, Q_2 \in \mathcal{Q}_\Sigma$,

$$\begin{aligned}
\langle Q_1, (\mathrm{Sym}^2 S_e) Q_1 \rangle &= \mathrm{Tr}(Q_1 S_e Q_1 S_e) = \frac{1}{2}(\alpha_e^2 - 2\beta_e^2 + \gamma_e^2) = (\alpha_e\gamma_e - \beta_e^2) + \left(\frac{\alpha_e - \gamma_e}{\sqrt{2}}\right)^2, \\
\langle Q_2, (\mathrm{Sym}^2 S_e) Q_2 \rangle &= \mathrm{Tr}(Q_2 S_e Q_2 S_e) = \alpha_e\gamma_e + \beta_e^2 = (\alpha_e\gamma_e - \beta_e^2) + (\sqrt{2}\beta_e)^2, \\
\langle Q_1, (\mathrm{Sym}^2 S_e) Q_2 \rangle &= \mathrm{Tr}(Q_1 S_e Q_2 S_e) = \beta_e(\alpha_e - \gamma_e) = \left(\frac{\alpha_e - \gamma_e}{\sqrt{2}}\right)(\sqrt{2}\beta_e).
\end{aligned}$$

Summing over $e = 1, \ldots, m$ yields (39). Since $\dim(\mathcal{Q}_\Sigma) = 2$, Cauchy's Interlacing Theorem gives $\lambda_2(\mathcal{M}_r^+) \geq \lambda_2(\mathcal{M}_r^+|_{\mathcal{Q}_\Sigma}) \geq \langle w_\Sigma, \mathcal{M}_r^- w_\Sigma \rangle$. □

5.2 The 4×4 Hodge–Eigenbasis Operator Identity and Unconditional Resolution for All Rank-≤ 3 Positive Semidefinite Host Matrices

When $r = 3$, the exterior square $\wedge^2(\mathbb{R}^3)$ has dimension $\binom{3}{2} = 3$, and the Hodge star operator $\star : \wedge^2(\mathbb{R}^3) \xrightarrow{\sim} \mathbb{R}^3$,

$$\star(e_2 \wedge e_3) = e_1, \qquad \star(e_1 \wedge e_3) = -e_2, \qquad \star(e_1 \wedge e_2) = e_3,$$

is an $\mathrm{SO}(3)$-equivariant isometry: for every $Q \in \mathrm{SO}(3)$ and every $S \in \mathrm{Sym}_3(\mathbb{R})$, $\wedge^2(Q^T S Q) = (\wedge^2 Q)^T (\wedge^2 S)(\wedge^2 Q)$. Crucially, this means that *every* bivector in $\wedge^2(\mathbb{R}^3)$ is simple, and rotating $\mathbb{R}^3$ to the orthonormal eigenbasis of $\star\mathcal{M}_3^-\star^{-1}$ simultaneously diagonalizes $\mathcal{M}_3^- = \sum_{e=1}^m \wedge^2(S_e)$! Using this $\mathrm{SO}(3)$ Hodge duality, we discover an exact 4×4 operator identity that proves *pointwise* eigenvalue dominance $\lambda_{k+1}(\mathcal{M}_3^+) \geq \lambda_k(\mathcal{M}_3^-)$ for all $k = 1, 2, 3$.

Theorem 5.2 (Pointwise Eigenvalue Dominance for All Rank-≤ 3 Positive Semidefinite Host Matrices of Arbitrary Size $n \times n$)**.** *Let $S_1, \ldots, S_m \in \mathrm{Sym}_3(\mathbb{R})$ be any real symmetric 3×3 matrices ($m \geq 1$), and set*

$$\mathcal{M}_3^+ := \sum_{e=1}^m \mathrm{Sym}^2(S_e) \in \mathrm{Sym}_6(\mathbb{R}), \qquad \mathcal{M}_3^- := \sum_{e=1}^m \wedge^2(S_e) \in \mathrm{Sym}_3(\mathbb{R}).$$

(a) ***Single-Matrix*** 4×4 ***Cofactor–Rank-***1 ***Decomposition on*** U_4***:*** *Let $\{e_1, e_2, e_3\}$ be any orthonormal basis of $\mathbb{R}^3$, and define the 4-dimensional orthonormal subspace of traceless symmetric matrices with $(1,1)$-entry zero:*

$$U_4 := \{V \in \mathrm{Sym}_3(\mathbb{R}) : \mathrm{Tr}(V) = 0,\ V_{11} = 0\} = \mathrm{span}\{u_1, u_2, u_3, u_4\}, \tag{41}$$

where $u_1 := \frac{E_{22} - E_{33}}{\sqrt{2}}$, $u_2 := E_{23}^+$, $u_3 := E_{13}^+$, and $u_4 := E_{12}^+$. For any $S = (s_{ab})_{a,b=1}^3 \in \mathrm{Sym}_3(\mathbb{R})$, let $W := \wedge^2(S)$ be its 3×3 second compound matrix in the basis $\{e_2 \wedge e_3,\ e_1 \wedge e_3,\ e_1 \wedge e_2\}$:

$$W_{11} = s_{22}s_{33} - s_{23}^2, \quad W_{22} = s_{11}s_{33} - s_{13}^2, \quad W_{33} = s_{11}s_{22} - s_{12}^2,$$

$$W_{12} = s_{21}s_{33} - s_{23}s_{13}, \quad W_{13} = s_{21}s_{32} - s_{22}s_{13}, \quad W_{23} = s_{11}s_{32} - s_{12}s_{13}.$$

Then the 4×4 compression of $\mathrm{Sym}^2(S)$ to U_4 satisfies the exact polynomial identity

$$\boxed{\mathrm{Sym}^2(S)\big|_{U_4} = \begin{pmatrix} W_{11} & 0 & W_{13} & -W_{12} \\ 0 & W_{11} & W_{12} & -W_{13} \\ W_{13} & W_{12} & W_{22} & W_{23} \\ -W_{12} & -W_{13} & W_{23} & W_{33} \end{pmatrix} + \mathbf{z}_S \mathbf{z}_S^T, \qquad \mathbf{z}_S := \begin{pmatrix} \frac{s_{22} - s_{33}}{\sqrt{2}} \\ \sqrt{2}\, s_{23} \\ \sqrt{2}\, s_{13} \\ \sqrt{2}\, s_{12} \end{pmatrix}.} \tag{42}$$

(b) ***Exact Diagonalization in the Hodge Eigenbasis of $\mathcal{M}_3^-$:*** *Choose the orthonormal basis $\{e_1, e_2, e_3\}$ of $\mathbb{R}^3$ so that $\{e_2 \wedge e_3,\, e_1 \wedge e_3,\, e_1 \wedge e_2\}$ is an orthonormal eigenbasis of $\mathcal{M}_3^-$ with ordered eigenvalues $\tau_1 \geq \tau_2 \geq \tau_3$, i.e.,*

$$\mathcal{M}_3^- \;=\; \sum_{e=1}^{m} W^{(e)} \;=\; \operatorname{diag}(\tau_1,\, \tau_2,\, \tau_3).$$

Then all off-diagonal cofactor sums vanish ($\sum_{e=1}^m W_{ab}^{(e)} = 0$ for $a \neq b$), and summing (42) over $e = 1, \ldots, m$ yields the exact 4×4 positive-semidefinite perturbation identity

$$\boxed{\mathcal{M}_3^+|_{U_4} \;=\; \operatorname{diag}(\tau_1,\, \tau_1,\, \tau_2,\, \tau_3) \;+\; \sum_{e=1}^{m} \mathbf{z}_{S_e} \mathbf{z}_{S_e}^T \;\succeq\; \operatorname{diag}(\tau_1,\, \tau_1,\, \tau_2,\, \tau_3).} \tag{43}$$

(c) ***Pointwise Eigenvalue Dominance:*** *The eigenvalues $\lambda_1(\mathcal{M}_3^+) \geq \cdots \geq \lambda_6(\mathcal{M}_3^+)$ and $\lambda_1(\mathcal{M}_3^-) \geq \lambda_2(\mathcal{M}_3^-) \geq \lambda_3(\mathcal{M}_3^-)$ satisfy the pointwise inequalities*

$$\lambda_2(\mathcal{M}_3^+) \;\geq\; \lambda_1(\mathcal{M}_3^-), \qquad \lambda_3(\mathcal{M}_3^+) \;\geq\; \lambda_2(\mathcal{M}_3^-), \qquad \lambda_4(\mathcal{M}_3^+) \;\geq\; \lambda_3(\mathcal{M}_3^-). \tag{44}$$

(d) ***Unconditional Resolution of Rossman's Conjectures for All Rank-≤ 3 PSD Host Matrices:*** *For every nonnegative symmetric positive semidefinite matrix $G \in \mathbb{R}^{n \times n}_{\geq 0}$ of rank $r \leq 3$ and arbitrary dimension $n \geq 1$, the eigenvalues $\widetilde{\lambda}_1 \geq \widetilde{\lambda}_2 \geq \cdots \geq \widetilde{\lambda}_{n^2}$ of $\widetilde{G}$ satisfy*

$$\widetilde{\lambda}_2 + \widetilde{\lambda}_{n^2} \;\geq\; 0, \qquad \widetilde{\lambda}_3 + \widetilde{\lambda}_{n^2-1} \;\geq\; 0, \qquad \widetilde{\lambda}_4 + \widetilde{\lambda}_{n^2-2} \;\geq\; 0, \tag{45}$$

with $\widetilde{\lambda}_k \geq 0$ for all $1 \leq k \leq n^2 - 3$. Consequently, Rossman's Majorization Conjecture $(**)$, Conjecture 3.2 (pseudo-positivity of $\widetilde{G}$), and Sidorenko's inequality (1) for $K_{5,5} \setminus C_{10}$ and all bipartite Möbius ladders ML_{2m} hold unconditionally for all positive semidefinite host matrices $G \succeq 0$ of rank ≤ 3.

Proof. (a) In the standard orthonormal basis $(E_{11}, E_{22}, E_{33}, E_{23}^+, E_{13}^+, E_{12}^+)$ of $\operatorname{Sym}_3(\mathbb{R})$, the 4-dimensional subspace U_4 is spanned by the orthonormal vectors $u_1 = \frac{1}{\sqrt{2}}(E_{22} - E_{33})$, $u_2 = E_{23}^+$, $u_3 = E_{13}^+$, $u_4 = E_{12}^+$. Notice first that $\{u_1, u_2\} = \mathcal{Q}_{\operatorname{span}\{e_2, e_3\}}$, so the upper-left 2×2 block of $\operatorname{Sym}^2(S)|_{U_4}$ is given immediately by Lemma 5.1 with $\alpha = s_{22}, \beta = s_{23}, \gamma = s_{33}$:

$$\begin{pmatrix} \langle u_1, (\operatorname{Sym}^2 S) u_1 \rangle & \langle u_1, (\operatorname{Sym}^2 S) u_2 \rangle \\ \langle u_2, (\operatorname{Sym}^2 S) u_1 \rangle & \langle u_2, (\operatorname{Sym}^2 S) u_2 \rangle \end{pmatrix} \;=\; W_{11} I_2 + \begin{pmatrix} (s_{22} - s_{33})/\sqrt{2} \\ \sqrt{2} s_{23} \end{pmatrix} \begin{pmatrix} (s_{22} - s_{33})/\sqrt{2} & \sqrt{2} s_{23} \end{pmatrix}.$$

For the remaining entries on $\{u_3, u_4\}$ and the cross-block between $\{u_1, u_2\}$ and $\{u_3, u_4\}$, direct evaluation of $\langle u_a, (\operatorname{Sym}^2 S) u_b \rangle = \operatorname{Tr}(u_a S u_b S)$ using (9)–(11) (with $t = 0, B = S$) yields:

$$\begin{aligned}
\langle u_3, (\operatorname{Sym}^2 S) u_3 \rangle &= s_{11} s_{33} + s_{13}^2 = (s_{11} s_{33} - s_{13}^2) + 2 s_{13}^2 = W_{22} + (\mathbf{z}_S)_3^2, \\
\langle u_4, (\operatorname{Sym}^2 S) u_4 \rangle &= s_{11} s_{22} + s_{12}^2 = (s_{11} s_{22} - s_{12}^2) + 2 s_{12}^2 = W_{33} + (\mathbf{z}_S)_4^2, \\
\langle u_3, (\operatorname{Sym}^2 S) u_4 \rangle &= s_{11} s_{23} + s_{12} s_{13} = (s_{11} s_{23} - s_{12} s_{13}) + 2 s_{12} s_{13} = W_{23} + (\mathbf{z}_S)_3 (\mathbf{z}_S)_4, \\
\langle u_1, (\operatorname{Sym}^2 S) u_3 \rangle &= s_{12} s_{23} - s_{13} s_{33} = (s_{12} s_{23} - s_{22} s_{13}) + (s_{22} - s_{33}) s_{13} = W_{13} + (\mathbf{z}_S)_1 (\mathbf{z}_S)_3, \\
\langle u_1, (\operatorname{Sym}^2 S) u_4 \rangle &= s_{12} s_{22} - s_{13} s_{23} = -(s_{12} s_{33} - s_{23} s_{13}) + (s_{22} - s_{33}) s_{12} = -W_{12} + (\mathbf{z}_S)_1 (\mathbf{z}_S)_4, \\
\langle u_2, (\operatorname{Sym}^2 S) u_3 \rangle &= s_{12} s_{33} + s_{13} s_{23} = (s_{12} s_{33} - s_{23} s_{13}) + 2 s_{23} s_{13} = W_{12} + (\mathbf{z}_S)_2 (\mathbf{z}_S)_3, \\
\langle u_2, (\operatorname{Sym}^2 S) u_4 \rangle &= s_{13} s_{22} + s_{12} s_{23} = -(s_{12} s_{23} - s_{22} s_{13}) + 2 s_{23} s_{12} = -W_{13} + (\mathbf{z}_S)_2 (\mathbf{z}_S)_4.
\end{aligned}$$

This establishes the exact matrix identity (42) (formally verified in Lean 4 as Theorem 5B).

(b) Since the Hodge star $\star : \wedge^2(\mathbb{R}^3) \xrightarrow{\sim} \mathbb{R}^3$ is SO(3)-equivariant, choosing $\{e_1, e_2, e_3\}$ to be an orthonormal eigenbasis of $\star \mathcal{M}_3^- \star^{-1}$ makes $\{e_2 \wedge e_3, e_1 \wedge e_3, e_1 \wedge e_2\}$ an orthonormal

eigenbasis of $\mathcal{M}_3^-$ with eigenvalues $\tau_1 \geq \tau_2 \geq \tau_3$. In this basis, $\sum_{e=1}^m W_{11}^{(e)} = \tau_1$, $\sum_{e=1}^m W_{22}^{(e)} = \tau_2$, $\sum_{e=1}^m W_{33}^{(e)} = \tau_3$, and $\sum_{e=1}^m W_{ab}^{(e)} = 0$ for all $1 \leq a < b \leq 3$. Summing (42) over $e = 1, \ldots, m$ immediately gives (43).

(c) Let $D_4 := \mathrm{diag}(\tau_1, \tau_1, \tau_2, \tau_3)$, whose ordered eigenvalues are $\lambda_1(D_4) = \tau_1$, $\lambda_2(D_4) = \tau_1$, $\lambda_3(D_4) = \tau_2$, $\lambda_4(D_4) = \tau_3$. Since $\sum_{e=1}^m \mathbf{z}_{S_e}\mathbf{z}_{S_e}^T \succeq 0$, Weyl's Monotonicity Theorem applied to (43) yields $\lambda_j(\mathcal{M}_3^+|_{U_4}) \geq \lambda_j(D_4)$ for $j = 1, 2, 3, 4$. Meanwhile, since $U_4 \subset \mathrm{Sym}_3(\mathbb{R})$ is a 4-dimensional subspace of a 6-dimensional space, Cauchy's Interlacing Theorem gives $\lambda_j(\mathcal{M}_3^+) \geq \lambda_j(\mathcal{M}_3^+|_{U_4})$ for $j = 1, 2, 3, 4$. Combining these inequalities for $j = 2, 3, 4$ yields

$$\begin{aligned}
\lambda_2(\mathcal{M}_3^+) \;&\geq\; \lambda_2(\mathcal{M}_3^+|_{U_4}) \;\geq\; \lambda_2(D_4) \;=\; \tau_1 \;=\; \lambda_1(\mathcal{M}_3^-),\\
\lambda_3(\mathcal{M}_3^+) \;&\geq\; \lambda_3(\mathcal{M}_3^+|_{U_4}) \;\geq\; \lambda_3(D_4) \;=\; \tau_2 \;=\; \lambda_2(\mathcal{M}_3^-),\\
\lambda_4(\mathcal{M}_3^+) \;&\geq\; \lambda_4(\mathcal{M}_3^+|_{U_4}) \;\geq\; \lambda_4(D_4) \;=\; \tau_3 \;=\; \lambda_3(\mathcal{M}_3^-),
\end{aligned}$$

proving (44).

(d) For $G \succeq 0$ of rank $r \leq 3$ (padding with zero factors if $r < 3$), Theorem 3.1 shows that the nonzero spectrum of $\widetilde{G}$ is that of $\mathcal{M}_3^+ \oplus (-\mathcal{M}_3^-)$, where $\mathcal{M}_3^+ \succeq 0$ and $\mathcal{M}_3^- \succeq 0$. Thus $\widetilde{G}$ has at most three negative eigenvalues $\widetilde{\lambda}_{n^2} = -\lambda_1(\mathcal{M}_3^-) \leq \widetilde{\lambda}_{n^2-1} = -\lambda_2(\mathcal{M}_3^-) \leq \widetilde{\lambda}_{n^2-2} = -\lambda_3(\mathcal{M}_3^-) \leq 0$, while $\widetilde{\lambda}_k = \lambda_k(\mathcal{M}_3^+)$ for $k = 1, 2, 3, 4$. Hence (44) directly implies (45), which implies Rossman's Majorization Conjecture $(**)$ for all $k \in \{2, \ldots, n^2\}$, pseudo-positivity of $\widetilde{G}$ (Conjecture 3.2) by Karamata's inequality, and Sidorenko's inequality (1) for $K_{5,5} \setminus C_{10}$ and all ML_{2m}. □

Remark 5.3 (Hierarchy of Exterior–Symmetric Spectral Dominance in Higher Rank $r \geq 4$). Theorem 5.2 reveals a striking algebraic hierarchy for the exterior–symmetric dominance inequality $\lambda_{k+1}(\sum_{e=1}^m \mathrm{Sym}^2(S_e)) \geq \lambda_k(\sum_{e=1}^m \wedge^2(S_e))$:

(i) **Arbitrary Symmetric Matrices** $S_1, \ldots, S_m \in \mathrm{Sym}_r(\mathbb{R})$**:** Holds unconditionally for $r = 2$ (Theorem 4.3) and $r = 3$ (Theorem 5.2), where every eigenvector of $\mathcal{M}_r^-$ is a simple bivector. For $r \geq 4$, $\wedge^2(\mathbb{R}^r)$ contains non-decomposable bivectors (e.g. $e_1 \wedge e_2 + e_3 \wedge e_4$), and the inequality can fail for unrestricted S_e if $(S_e)_{ab}$ is not symmetric under permutation of (e, a, b).

(ii) **Totally Symmetric Cubic Tensors** $T_{eab} = (S_e)_{ab} \in \mathrm{Sym}^3(\mathbb{R}^r)$**:** Because the cubic moment matrices $(S_e)_{ab} = \sum_{i=1}^n x_i^{(e)} x_i^{(a)} x_i^{(b)}$ satisfy the full permutation symmetry $(S_e)_{ab} = (S_a)_{eb} = (S_b)_{ea}$, the family of cubic moment tensors is strictly constrained. Extensive numerical experiments indicate that for general $G \succeq 0$ of rank $r \geq 4$ (where $\mathcal{M}_r^+$ and $\mathcal{M}_r^-$ have the moment structure (38)), the pointwise dominance $\lambda_{k+1}(\mathcal{M}_r^+) \geq \lambda_k(\mathcal{M}_r^-)$ continues to hold across all $1 \leq k \leq \binom{r}{2}$.

5.3 Exact 6×6 Cofactor Representation and Zero-Diagonal 3×3 Indefinite Matrices Across All $t \in [0, 1]$

For $n = 3$ and arbitrary $t \in [0, 1]$, Theorem 2.1(c) and Theorem 2.3 combine into a remarkably clean explicit representation of the entire 6×6 deficit matrix $\mathcal{L}_+(t) - ww^T$ in terms of the cofactor matrix $Z := \wedge^2(B)$ ($B = A^{\circ(1-t)}$). Specifically, letting $Z_{i,j;k,l} := B_{ik}B_{jl} - B_{il}B_{jk}$ be the 2×2 minors of B, every entry of $\mathcal{L}_+(t) - ww^T$ on $\mathcal{B}_+ = (E_{11}, E_{22}, E_{33}, E_{12}^+, E_{13}^+, E_{23}^+)$ is a single minor of B (or a sum of two symmetric minors on the off-diagonal diagonal entries):

$$\begin{aligned}
(\mathcal{L}_+(t) - ww^T)_{(i,i),(k,k)} \;&=\; -A_{ii}^t A_{kk}^t \, Z_{i,k;i,k}, \qquad (\mathcal{L}_+(t) - ww^T)_{(i,i),(i,j)} \;=\; 0, && (46)\\
(\mathcal{L}_+(t) - ww^T)_{(i,i),(j,k)} \;&=\; \sqrt{2}\, A_{ii}^t A_{jk}^t \, Z_{i,j;i,k} \quad (i \notin \{j, k\}), && (47)\\
(\mathcal{L}_+(t) - ww^T)_{(i,j),(i,k)} \;&=\; A_{ij}^t A_{ik}^t \, Z_{i,j;i,k} \;=\; \mathcal{L}_-(t)_{(i,j),(i,k)} \quad (j \neq k), && (48)\\
(\mathcal{L}_+(t) - ww^T)_{(i,j),(i,j)} \;&=\; 2A_{ij}^{2t}(B_{ii}B_{jj} - B_{ij}^2) - \mathcal{L}_-(t)_{(i,j),(i,j)}. && (49)
\end{aligned}$$

We conclude this section by proving unconditional eigenvalue dominance across *all* $t \in [0,1]$ for the fundamental indefinite rank-3 family of zero-diagonal 3×3 matrices (weighted triangles).

Theorem 5.4 (Unconditional Eigenvalue Dominance for Zero-Diagonal 3×3 Indefinite Matrices Across All $t \in [0,1]$)**.** *Let* $A = \begin{pmatrix} 0 & c_3 & c_2 \\ c_3 & 0 & c_1 \\ c_2 & c_1 & 0 \end{pmatrix}$ *be any zero-diagonal* 3×3 *nonnegative symmetric matrix* ($c_1, c_2, c_3 \geq 0$)*, which has rank* 3 *and indefinite signature* $(1,2)$ *whenever* $c_1, c_2, c_3 > 0$*. For any* $t \in [0,1]$*, set* $x_1 := c_1^2, x_2 := c_2^2, x_3 := c_3^2$ *ordered as* $x_1 \geq x_2 \geq x_3 \geq 0$*, and let* $u := (\sqrt{x_1}, \sqrt{x_2}, \sqrt{x_3})^T \in \mathbb{R}^3$*. Then:*

(a) ***Exact Rank-1 and Rank-1-Perturbed Diagonal Structure:*** *For every* $t \in [0,1]$*,* $M(t)$ *is independent of* t*:* $\mathcal{L}_+(t)$ *vanishes on* $\operatorname{span}\{E_{11}, E_{22}, E_{33}\}$ *and acts on* $\operatorname{span}\{E_{23}^+, E_{13}^+, E_{12}^+\}$ *as the rank-1 positive semidefinite matrix* $\mathcal{L}_+^{\text{off}}(t) = uu^T$ *with sole nonzero eigenvalue* $\lambda_1(\mathcal{L}_+(t)) = x_1 + x_2 + x_3$*. In the signed cyclic skew basis* $(E_{23}^-, -E_{13}^-, E_{12}^-)$*, the negative-sector operator is*

$$-\mathcal{L}_-(t) \;=\; 2\,\mathrm{diag}(x_1,\, x_2,\, x_3) \;-\; uu^T. \tag{50}$$

(b) ***Unconditional Eigenvalue Dominance and Majorization:*** *Let* $\mu_1 \geq \mu_2 \geq \mu_3$ *be the eigenvalues of* $-\mathcal{L}_-(t)$*. Then* $\mu_1 \geq \mu_2 \geq 0 \geq \mu_3$*, and thus* $\lambda_2(M(t)) = \mu_1$*,* $\lambda_3(M(t)) = \mu_2$*, and* $\lambda_9(M(t)) = \mu_3$*, with all other eigenvalues besides* $\lambda_1(M(t)) = x_1 + x_2 + x_3$ *equal to* 0*. Moreover,*

$$\boxed{\lambda_2(M(t)) + \lambda_9(M(t)) \;=\; \mu_1 + \mu_3 \;=\; (x_1 + x_2 + x_3) - \mu_2 \;\geq\; (x_1 - x_2) + x_3 \;\geq\; 0.} \tag{51}$$

Consequently, Rossman's Conjecture 3.2, Conjecture 4.2 (for all $t \in [0,1]$*), the Majorization Conjecture* $(**)$*, and Sidorenko's inequality for* $K_{5,5} \setminus C_{10}$ *and all* ML_{2m} *hold unconditionally for all zero-diagonal* 3×3 *nonnegative symmetric matrices.*

Proof. (a) Since $A_{11} = A_{22} = A_{33} = 0$, every entry of $M(t)$ involving a diagonal pair (i,i) vanishes. On the off-diagonal pairs $\{(2,3),(1,3),(1,2)\}$ (indexed by $r \in \{1,2,3\}$ with edge weight $c_r = \sqrt{x_r}$), equations (11)–(12) yield $\mathcal{L}_+(t)_{(i,j),(i,j)} = A_{ij}^{2t}(0 + B_{ij}^2) = A_{ij}^2 = x_r$ and $\mathcal{L}_-(t)_{(i,j),(i,j)} = A_{ij}^{2t}(0 - B_{ij}^2) = -x_r$. For two distinct pairs sharing a vertex, say $(1,2)$ and $(1,3)$, $B_{11}B_{23} \pm B_{13}B_{21} = \pm B_{13}B_{12}$, so $\mathcal{L}_+(t)_{(1,2),(1,3)} = c_3c_2 = \sqrt{x_3x_2}$ and $\mathcal{L}_-(t)_{(1,2),(1,3)} = -c_3c_2 = -\sqrt{x_3x_2}$. Thus $\mathcal{L}_+^{\text{off}}(t) = uu^T$ with sole nonzero eigenvalue $\|u\|^2 = x_1 + x_2 + x_3$. In the signed cyclic basis $(E_{23}^-, -E_{13}^-, E_{12}^-)$, the sign flip on E_{13}^- makes all off-diagonal entries of $\mathcal{L}_-(t)$ equal to $+\sqrt{x_rx_s}$ $(r \neq s)$ while the diagonal entries are $-x_r = u_r^2 - 2x_r$, so $\mathcal{L}_-(t) = uu^T - 2\,\mathrm{diag}(x_1,x_2,x_3)$, proving (50).

(b) Since $-\mathcal{L}_-(t) = 2\,\mathrm{diag}(x_1,x_2,x_3) - uu^T$ is a rank-1 negative perturbation of $D := 2\,\mathrm{diag}(x_1,x_2,x_3)$ with eigenvalues $2x_1 \geq 2x_2 \geq 2x_3 \geq 0$, Weyl's Interlacing Theorem implies

$$2x_1 \;\geq\; \mu_1 \;\geq\; 2x_2 \;\geq\; \mu_2 \;\geq\; 2x_3 \;\geq\; \mu_3.$$

In particular, $\mu_1 \geq \mu_2 \geq 0$ and $\mu_2 \leq 2x_2$. Moreover,

$$\mu_1 + \mu_2 + \mu_3 \;=\; \mathrm{Tr}\,(-\mathcal{L}_-(t)) \;=\; 2(x_1 + x_2 + x_3) - (x_1 + x_2 + x_3) \;=\; x_1 + x_2 + x_3.$$

Therefore,

$$\mu_1 + \mu_3 \;=\; (x_1 + x_2 + x_3) - \mu_2 \;\geq\; (x_1 + x_2 + x_3) - 2x_2 \;=\; (x_1 - x_2) + x_3 \;\geq\; 0,$$

which proves (51) (formally verified in Lean 4 as `zero_diag_3x3_dominance`). □

6 Disproof of Rossman's General Conjecture 4.2 for $n \geq 3$ and $t \in (2/3, 1)$

In both the abstract and Section 4 of [4], Rossman conjectured (Conjecture 4.2) that for every positive symmetric $n \times n$ matrix A, the matrix $M(t)$ has a pseudo-positive spectrum for *all* $t \in [0, 1]$, noting that $M(t)$ is visibly pseudo-positive at the endpoints $t = 0, 1/2, 1$. Our sixth main theorem gives an explicit analytic disproof of Conjecture 4.2 for all $n \geq 3$ and all $t \in (2/3, 1)$, and explains why the parameter $t = 1/3$ ($\widetilde{G}$) lies inside the stable regime.[1]

Theorem 6.1 (Analytic Disproof of Rossman's Conjecture 4.2 for $n \geq 3$ and $t \in (2/3, 1)$)**.** *Let $n \geq 3$ and consider the 3×3 symmetric matrix family*

$$A_\varepsilon \;=\; \begin{pmatrix} a & b & c \\ b & d & \varepsilon \\ c & \varepsilon & \varepsilon \end{pmatrix}, \qquad a, b, c, d > 0, \quad \varepsilon \to 0^+. \tag{52}$$

Let $r := (ad)^t (b^2)^{1-t}$ and $s := (b^2)^t (ad)^{1-t}$.

(a) ***Exact Block Reduction as $\varepsilon \to 0^+$:*** *In the limit $\varepsilon \to 0^+$, $M(t) = \mathcal{L}_+(t) \oplus (-\mathcal{L}_-(t))$ has three zero eigenvalues (on coordinates $(3,3)$ and $(2,3)^\pm$), while its remaining 6 eigenvalues are the eigenvalues of the 4×4 symmetric block $\mathcal{L}_{+,4}(t)$ on $\{(1,1), (2,2), (1,2)^+, (1,3)^+\}$ and the 2×2 block $-\mathcal{L}_{-,2}(t)$ on $\{(1,2)^-, (1,3)^-\}$:*

$$\mathcal{L}_{+,4}(t) \;=\; \begin{pmatrix} a^2 & r & \sqrt{2}ab & \sqrt{2}ac \\ r & d^2 & \sqrt{2}bd & 0 \\ \sqrt{2}ab & \sqrt{2}bd & s + b^2 & bc \\ \sqrt{2}ac & 0 & bc & c^2 \end{pmatrix}, \qquad -\mathcal{L}_{-,2}(t) \;=\; \begin{pmatrix} b^2 - s & bc \\ bc & c^2 \end{pmatrix}. \tag{53}$$

(b) ***Exact Principal Submatrix Determinant Identity:*** *Let $L_{124}(t)$ be the 3×3 principal submatrix of $\mathcal{L}_{+,4}(t)$ on indices $\{1, 2, 4\}$ (with $a = 1$). Then at $\lambda = -c^2$,*

$$\boxed{\det\left(L_{124}(t) + c^2 I_3\right) \;=\; 2c^2\Big(c^2(d^2 + c^2) - r^2\Big).} \tag{54}$$

In particular, whenever $r^2 > c^2(d^2 + c^2)$, the minimum eigenvalue of $\mathcal{L}_{+,4}(t)$ satisfies $\lambda_4(\mathcal{L}_{+,4}(t)) \leq \lambda_{\min}(L_{124}(t)) < -c^2$.

(c) ***Analytic Counterexample Curve for All $t \in (2/3, 1)$:*** *For every $t \in (2/3, 1)$, the interval $(\frac{2t}{2t-1}, \frac{2-t}{1-t})$ is nonempty (since $\frac{2t}{2t-1} < \frac{2-t}{1-t} \iff 3t > 2$). Choosing any $\alpha \in (\frac{2t}{2t-1}, \frac{2-t}{1-t})$, $\kappa > 1$, and setting $a = 1$, $d = \eta$, $c = \kappa\eta$, $b = \eta^{\alpha/2}$ as $\eta \to 0^+$, the eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_9$ of $M(t)$ satisfy*

$$\lambda_2(M(t)) + \lambda_9(M(t)) \;=\; -\frac{2\kappa^2}{1 + \kappa^2}\, \eta^{2t + 2\alpha(1-t)}(1 + o(1)) \;<\; 0. \tag{55}$$

Consequently, $|\lambda_9(M(t))| > \lambda_2(M(t)) = \max_{2 \leq k \leq 8} |\lambda_k(M(t))|$, so for all sufficiently large p, $\sum_{k=2}^{9} \operatorname{sign}(\lambda_k)|\lambda_k|^p < 0$, which forces $f_t(p) < \lambda_1(M(t)) = \lim_{q \to \infty} f_t(q)$ and implies that $f_t(p)$ is strictly increasing on a neighborhood of ∞, disproving Rossman's Conjecture 4.2.

[1] In personal communication [5], Rossman noted that numerical counterexamples to Conjecture 4.2 for $t > 2/3$ were also independently found via computer-assisted experiments after [4] was posted; Theorem 6.1 provides the exact analytic mechanism and determinant threshold.

(d) ***Explicit Numerical Counterexample at*** $t = 0.8$***:*** *For the strictly positive symmetric matrix*

$$A = \begin{pmatrix} 1.00 & 0.08 & 0.16 \\ 0.08 & 0.12 & 10^{-12} \\ 0.16 & 10^{-12} & 10^{-12} \end{pmatrix}$$

at $t = 0.8 \in (2/3, 1)$*, the nonzero eigenvalues of* $M(0.8)$ *are:*

$$\lambda_1 \approx 1.066439, \qquad \lambda_2 \approx 0.030236, \qquad \lambda_3 \approx 0.022148,$$
$$\lambda_4 \approx 0.001111, \qquad \lambda_8 \approx -0.009738, \qquad \lambda_9 \approx -0.031797,$$

with $\lambda_5, \lambda_6, \lambda_7 \approx 0$*. Here* $\lambda_2 + \lambda_9 \approx -0.001561 < 0$*, the tail sum* $\sum_{k=2}^{9} \operatorname{sign}(\lambda_k)|\lambda_k|^p$ *is strictly negative for all* $p \geq 5$ *(at* $p = 5$*,* $\sum_{k=2}^{9} \operatorname{sign}(\lambda_k)|\lambda_k|^5 \approx -1.987 \times 10^{-9} < 0$*), and* $f_{0.8}(5) < f_{0.8}(6) < \lambda_1$*. By contrast, at* $t = 1/3$ *(*$\widetilde{G}$*), the same matrix* A *satisfies* $\widetilde{\lambda}_2 \approx 0.045672$*,* $\widetilde{\lambda}_9 \approx -0.041222$*,* $\widetilde{\lambda}_2 + \widetilde{\lambda}_9 \approx +0.004451 > 0$*, and* $f_{1/3}(p)$ *is strictly decreasing over* $p \in [1, \infty)$*.*

Proof. (a) Evaluating (9)–(12) on A_ε as $\varepsilon \to 0^+$ immediately gives (53).

(b) For $a = 1$, $L_{124}(t) + c^2 I_3 = \begin{pmatrix} 1+c^2 & r & \sqrt{2}c \\ r & d^2+c^2 & 0 \\ \sqrt{2}c & 0 & 2c^2 \end{pmatrix}$. Expanding the determinant along the third column gives

$$\det\left(L_{124}(t)+c^2 I_3\right) = -\sqrt{2}c(\sqrt{2}c(d^2+c^2)) + 2c^2\Big((1+c^2)(d^2+c^2) - r^2\Big) = 2c^2\Big(c^2(d^2+c^2) - r^2\Big),$$

proving (54). By Cauchy's Interlacing Theorem, $\lambda_4(\mathcal{L}_{+,4}(t)) \leq \lambda_3(L_{124}(t)) < -c^2$ whenever $r^2 > c^2(d^2 + c^2)$.

(c) Along $a = 1, d = \eta, c = \kappa\eta$ $(\kappa > 1)$, $b = \eta^{\alpha/2}$ with $\frac{2t}{2t-1} < \alpha < \frac{2-t}{1-t}$:

(i) Since $\alpha > \frac{2t}{2t-1} > \frac{1+t}{t}$, $s = \eta^{\alpha t+1-t} = o(\eta^2) = o(d^2)$ and $b^2 = \eta^\alpha = o(s)$.

(ii) In $-\mathcal{L}_{-,2}(t) = \begin{pmatrix} b^2-s & bc \\ bc & c^2 \end{pmatrix}$, the positive eigenvalue is $\lambda_1(-\mathcal{L}_{-,2}) = c^2 + \frac{b^2c^2}{c^2+s-b^2}(1 + o(1)) = c^2 + \Theta(\eta^\alpha)$.

(iii) In $\mathcal{L}_{+,4}(t)$, the three eigenvalues other than $\lambda_1(\mathcal{L}_{+,4}) = 1 + O(\eta^2)$ are $\lambda_2(\mathcal{L}_{+,4}) = d^2(1 + o(1)) = \eta^2(1 + o(1)) < \kappa^2\eta^2 = c^2$, $\lambda_3(\mathcal{L}_{+,4}) = s(1 + o(1)) = o(\eta^2)$, and by (54),

$$\lambda_4(\mathcal{L}_{+,4}) = -c^2 - \frac{2c^2(r^2 - c^2(d^2 + c^2))}{d^2 + c^2}(1 + o(1)).$$

Since $\alpha < \frac{2-t}{1-t}$, $2t + 2\alpha(1-t) < 4$, so $r^2 = \eta^{2t+2\alpha(1-t)} \gg \eta^4 \asymp c^2(d^2 + c^2)$. Since $\alpha > \frac{2t}{2t-1}$, $2t + 2\alpha(1-t) < \alpha$, so $r^2 \gg \eta^\alpha = b^2$. Thus $\lambda_2(M(t)) = \lambda_1(-\mathcal{L}_{-,2}) = c^2 + o(r^2)$ while $\lambda_9(M(t)) = \lambda_4(\mathcal{L}_{+,4}) = -c^2 - \frac{2\kappa^2}{1+\kappa^2}r^2(1 + o(1))$, giving (55).

□

Table 1: Comparison of the extremal eigenvalues of $M(t)$ for $A = \begin{pmatrix} 1.00 & 0.08 & 0.16 \\ 0.08 & 0.12 & 0 \\ 0.16 & 0 & 0 \end{pmatrix}$ across $t \in [1/3, 0.90]$, illustrating the phase transition where $|\lambda_9(M(t))|$ overtakes $\lambda_2(M(t))$ for $t > 2/3$.

t	r	s	$\lambda_2(\mathcal{L}_+)$	$\lambda_1(-\mathcal{L}_-)$	$\lambda_9(M(t))$	$\lambda_2(M(t)) + \lambda_9(M(t))$
$1/3 \approx 0.3333$	0.017000	0.045172	0.045672	0.028052	-0.041222	$+0.004451 > 0$
0.5000	0.027713	0.027713	0.031962	0.028865	-0.027529	$+0.004433 > 0$
0.6000	0.037149	0.020672	0.027313	0.029355	-0.028361	$+0.000994 > 0$
0.7000	0.049806	0.015420	0.024245	0.029819	-0.029638	$+0.000181 > 0$
0.8000	**0.066770**	**0.011502**	**0.022148**	**0.030236**	$\mathbf{-0.031797}$	$\mathbf{-0.001561 < 0}$
0.9000	**0.089512**	**0.008580**	**0.020438**	**0.030601**	$\mathbf{-0.035518}$	$\mathbf{-0.004917 < 0}$

Remark 6.2 (Why $t = 1/3$ ($\widetilde{G}$) Avoids the Counterexample). At $t = 1/3$, $r = (ad)^{1/3}(b^2)^{2/3}$ and $s = (b^2)^{1/3}(ad)^{2/3}$ satisfy the exact geometric progression relation $r^2 = sb^2$ and $s^2 = adr$. Consequently, whenever $s \ll d^2$ (as required in Step (i) of the counterexample construction), $r^2 = sb^2 \ll d^2b^2 \ll d^4 < c^2(d^2 + c^2)$, so by (54), $\det(L_{124}(1/3) + c^2 I_3) > 0$ and $|\lambda_4(\mathcal{L}_{+,4})|$ can never exceed c^2. Thus the cubic structure of $\widetilde{G} = M(1/3)$ enforces an exact coupling between r and s that blocks the $t > 2/3$ instability.

7 Formal Verification in Lean 4 and Computational Suite

All six theorems in this paper are accompanied by computer-verified artifacts:

(1) **Lean 4 Formalization (`SidorenkoMobiusTheorem.lean`):** Formalizes the universal off-diagonal minor and trace identity (Theorem 2.1(c)), the extended star-subspace decoupling identities (Theorem 2.3(b)), the $n = 2$ Muirhead factorization (26), the $\mathrm{SL}_2(\mathbb{R})$ Lorentzian identity (28), the exact Brahmagupta–Minkowski sum-of-squares proof of the Lorentzian Reverse Cauchy–Schwarz Inequality (37), the exact 4×4 Hodge–eigenbasis sum-of-squares identity (42)–(43) (Theorem 5.2), the zero-diagonal 3×3 indefinite dominance bound (51) (Theorem 5.4), and the 3×3 determinant identity (54).

(2) **Master Verification Suite (`verify_sidorenko_mobius.mjs`):** Verifies Theorems 2.1–6.1 across 523+ test configurations ($n = 2, \ldots, 12$, $t \in [0, 1]$, rank-≤ 2 PSD and indefinite matrices, rank-≤ 3 PSD host matrices of sizes $n = 3, \ldots, 12$, and zero-diagonal 3×3 indefinite matrices) with a 100% pass rate to machine precision ($< 10^{-11}$).

References

[1] D. Conlon, J. Fox, and B. Sudakov, *An approximate version of Sidorenko's conjecture*, Geom. Funct. Anal. **20** (2010), 1354–1366.

[2] D. Conlon, J. H. Kim, C. Lee, and J. Lee, *Some advances on Sidorenko's conjecture*, J. Lond. Math. Soc. (2) **98** (2018), 593–608.

[3] H. Hatami, *Graph norms and Sidorenko's conjecture*, Israel J. Math. **175** (2010), 125–150.

[4] B. Rossman, *On Sidorenko's Conjecture for Bipartite Möbius Ladders*, Duke University Preprint (May 8, 2025), 11 pp.

[5] B. Rossman, Personal communication (October 2026).

[6] A. F. Sidorenko, *Inequalities for functionals generated by bipartite graphs*, Diskret. Mat. **3** (1991), 50–65 (in Russian); Discrete Math. Appl. **2** (1992), 489–504.

[7] M. Simonovits, *Extremal graph problems, degenerate extremal problems, and supersaturated graphs*, in Progress in Graph Theory (Waterloo, Ont., 1982), Academic Press, Toronto, ON, 1984, pp. 333–358.

[8] B. Szegedy, *An information theoretic approach to Sidorenko's conjecture*, arXiv:1406.6738 (2015).